MILITARY SPECIFICATION

REGULATOR, VOLTAGE, GENERATOR: SOLID STATE 28 VOLT, 100 AMPERE, DIRECT CURRENT

This specification is approved for use by US Army Tank-Automotive Command, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers one type of voltage regulator of solid state design for 28 volt (V), 100 ampere (A), direct current (dc) engine electrical systems (see 6.1).

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. Unless otherwise specified, the following specifications, standards, and handbooks, of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation, form a part of this specification to the extent specified herein.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: US Army Tank-Automotive Command, ATTN: DRSTA-GSS, Warren, MI 48090, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document, or by letter.

FSC 2920

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:24Z
Check the source to verify that this is the current version before use.

SPECIFICATIONS
FEDERAL

QQ-S-365	- Silver Plating, Electrodeposited, General Requirements for.
QQ-Z-325	- Zinc Coating, Electrodeposited, Requirements for.
TT-C-490	- Cleaning Method and Pretreatment of Ferrous Surfaces for Organic Coatings.
TT-E-489	- Enamel, Alkyd, Gloss (for Exterior and Interior Surfaces).
TT-P-636	- Primer Coating, Alkyd, Wood and Ferrous Metal.
TT-P-1757	- Primer Coatings, Zinc Chromate, Low-Moisture-Sensitivity.
GGG-S-121	- Screwdriver and Screw Starter, Hand.

MILITARY

MIL-P-514	- Plate, Identification, Instruction and Marking, Blank.
MIL-C-5541	- Chemical Conversion Coatings, on Aluminum and Aluminum Alloys.
MIL-A-8625	- Anodic Coatings, for Aluminum and Aluminum Alloys.
MIL-T-12879	- Treatment, Chemical, Prepaint and Corrosion Inhibitive, for Zinc Surfaces.
MIL-T-13513	- Terminal, Lug, Splice, Conductor (Electrical, Solderless, for Automotive Use).
MIL-C-13516	- Circuit Breakers Manual & Automotive (28 Volt).
MIL-F-13927	- Fungus Resistance Test: Automotive Components.
MIL-F-15160	- Fuse, Instrument Power, and Telephone.

STANDARDS
FEDERAL

FED-STD-H28	- Screw Thread Standards for Federal Services.

MILITARY

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-130	- Identification Marking of US Military Property.
MIL-STD-202	- Test Methods for Electronic and Electrical Component Parts.
MIL-STD-454	- Standard General Requirements for Electronic Equipment.
MIL-STD-889	- Dissimilar Metals.

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:24Z
Check the source to verify that this is the currentversion before use.

MIL-STD-1184	- Electrical Components for Automotive Vehicles; Waterproofness Test.
MIL-STD-1275	- Electrical Circuit, 28 Volt DC Transient Characteristics for Military Vehicles.
MIL-STD-45662	- Calibration Systems Requirement.
MS3212	- Screw, Machine, Panhead, Self-Sealing U-Ring, Silicone Rubber, Integral.
MS35000	- Battery Storage, Lead-acid, Waterproof.

2.1.2 Other Government documents, Government drawings and publications. The following documents, drawings, and publications form a part of this specification to the extent specified herein.

DRAWINGS
ARMY

10947439	- Regulator, Voltage.
10947517	- Alternator Assembly.
7064429	- Connector.
7064706	- Connector.

(Copies of specifications, standards, drawings and publications required by contractors in connection with specific acquisition functions should be obtained from the acquisition activity, or as directed by the contracting officer).

2.2 Other publications. The following document(s) form a part of this specification to the extent specified herein. The issues of the documents which are indicated as DoD adopted shall be the issue listed in the current DoDISS and the supplement thereto, if applicable.

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM E-380	- Standard for Metric Practice.

(Application for copies should be addressed to American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA, 19103.)

(Industry association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 First article. Unless otherwise specified, the contractor shall furnish voltage regulators which shall be subjected to first article inspection (see 4.4 and 6.2). First article inspection samples, properly marked with identifying information shall be representative of the unit to be

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:24Z
Check the source to verify that this is the current version before use.

furnished to the Government. All subsequent voltage regulators delivered to the Government shall conform to these samples in all of their pertinent physical and performance attributes.

3.2 Materials. Materials shall be as specified herein and in referenced specifications, standards, and drawings. Materials shall be free of defects which adversely affect performance or serviceability of the finished product. Materials not specifically designated shall be suitable for use in a regulator operating over an ambient temperature range of -65 to +225 degrees Fahrenheit (°F) [-54 to +107 degrees Celsius (°C)] and storage over a temperature range of -80 to +225°F (-62 to +107°C), without any changes in physical or dimensional properties that would result in operation of the regulator falling outside the limits specified herein (see 4.6.1).

3.3 Design and construction. The regulator shall conform to Drawing 10947439, this specification and other applicable drawings, specifications and standards (see 4.5.2 and 4.6.2).

3.3.1 Voltage adjustment (see 3.4.1.10). Considering the surface with the four 10-32 holes as the regulator bottom, an access hole for voltage adjustment shall be provided on either the top surface or the connector end surface. The regulated voltage (see 6.4.4) shall be externally adjustable with a screwdriver whose tip conforms to type VI, class 1, style 1, bit number 1 of GSS-S-121. The access hole shall be sealed with a screw conforming to MS3212-47 (see 4.6.3).

3.3.2 Dissimilar metals. Dissimilar metals shall be protected from galvanic corrosion in accordance with requirements of MIL-STD-889 (see 4.6.1).

3.3.3 Standard parts. Military standard parts shall be incorporated whenever possible. Commercial standard parts may be used, provided they are interchangeable with military standard parts without modification (see 4.6.1).

3.3.4 Threaded parts. Screw threads of the form, number per inch, and class specified on Drawing 10947439 shall be in accordance with FED-STD-H28/2 (see 4.6.1).

3.3.5 Solder. Soldering shall conform to MIL-STD-454, requirement 5, flux type R. Composition of solder alloy shall be optional (see 4.6.1).

3.3.6 Polarity. The regulator shall operate in negative grounded vehicle electrical systems (see 4.6.1).

3.3.7 Electromagnetic interference suppression. Electromagnetic interference generated within the alternator, rectifier, and regulator shall be suppressed within the regulator housing (see 4.6.1).

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:24Z
Check the source to verify that this is the currentversion before use.

3.3.8 Finish.

3.3.8.1 Ferrous surfaces. Ferrous metal surfaces shall be cleaned and treated in accordance with type I or type III of TT-C-490. Exterior surfaces shall be given one coat of primer conforming to TT-P-636, 0.75 mil [0.02 millimeter (mm)] thick, or TT-P-1757, 0.40 to 0.60 mil (0.01 to 0.02 mm) thick (see 4.6.1).

3.3.8.2 Aluminum alloy surfaces. Aluminum surfaces shall be cleaned and treated in accordance with MIL-C-5541 or anodized in accordance with MIL-A-8625. Exterior surfaces shall be given one coat of primer conforming to TT-P-1757, 0.40 to 0.60 mil (0.01 to 0.02 mm) thick (see 4.6.1).

3.3.8.3 Electroplating. All unprotected metallic surfaces, (except those of corrosion-resistant metals or alloys), shall be zinc plated and shall conform to class 1 of QQ-Z-325. Silver plating, if specified, shall conform to QQ-S-365, with thickness of silver deposited as specified (see 4.6.1).

3.3.8.4 Zinc surfaces. Exterior zinc surfaces shall be cleaned and treated in accordance with MIL-T-12879, type I and given one coat of primer conforming to TT-P-1757, 0.40 to 0.60 mil (0.01 to 0.02 mm) thick (see 4.6.1).

3.3.8.5 Final paint. All exterior surfaces of the regulator assembly, except electrical connectors, mating surfaces on mounting brackets, and screw threads shall be painted. The finish coat shall be black gloss enamel conforming to TT-E-489, 0.75 to 1.25 mil (0.02 to 0.03 mm) thick (see 4.6.1).

3.3.9 Crimp terminations. Wire to terminal crimp connections shall conform to MIL-T-13513. Terminal material shall be optional (see 4.6.1).

3.3.10 Fuse or circuit breaker. If the regulator is equipped with a fuse or circuit breaker it shall conform, as applicable, to MIL-F-15160 or MIL-C-13516, type I. Design and construction shall be optional (see 4.6.1).

3.3.11 Overvoltage protection circuit. The circuitry necessary for sensing and reacting to an overvoltage condition (see 3.4.1.9) shall be different from the circuitry for establishing and regulating the regulated voltage. The only allowed common portion between the overvoltage and regulating circuits are wires, internal to the regulator, directly connected to the regulated voltage or the alternator output voltage (between pins C and D of connector J-2) (see 4.6.1).

3.4 Performance.

3.4.1 Electrical.

3.4.1.1 Voltage regulation. After being adjusted to the set point (see 6.4.6) the regulator shall establish and maintain a regulated voltage within 0.6 V dc of the nominal regulated voltage, as defined in 3.4.2.1 and shown for reference in figure 1, throughout the alternator speed range of 2000 to 8000 revolutions per minute (rpm) and, with load current from battery only to

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:24Z
Check the source to verify that this is the currentversion before use.

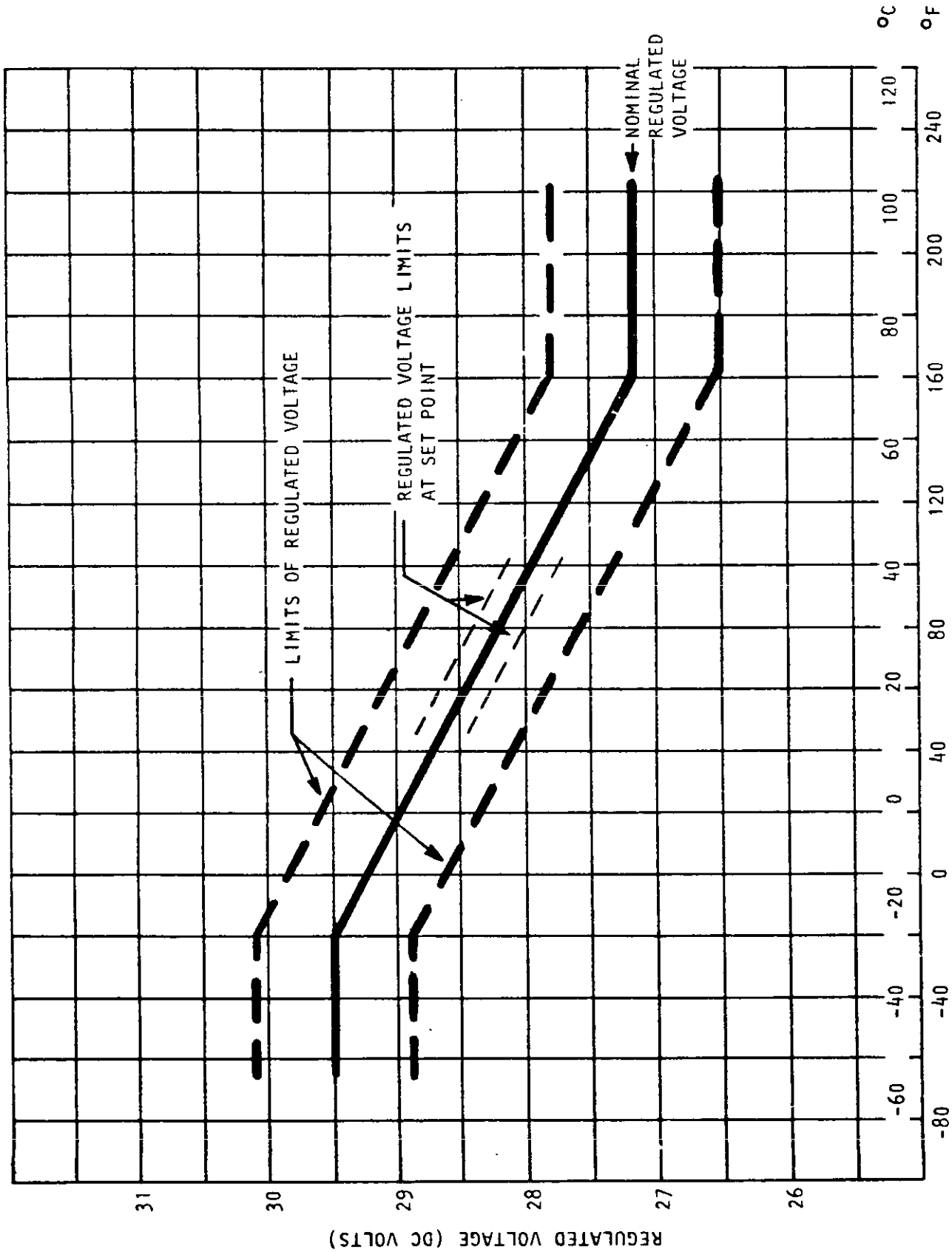


FIGURE 1. Ambient temperature influence on regulated voltage.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:24Z
Check the source to verify that this is the current version before use.

100 amperes (A). There shall be no sustained oscillation in the regulated voltage. After being subjected to the specified load changes, the regulator shall re-establish and maintain the regulated voltage within the time and voltage limits shown in figure 2 (see 4.6.4.1).

3.4.1.2 Low voltage. The regulator shall operate at any voltage above 16 V (see 4.6.4.2).

3.4.1.3 Transient and steady state voltage characteristics. Transient and steady state voltage shall conform to 3.4.1.3.1 through 3.4.1.3.4.

3.4.1.3.1 Ripple voltage. When the voltage regulator is operated in accordance with 3.4.1.1, the ripple voltage imposed on the regulated voltage shall be less than four volts peak to peak (see 4.6.4.3.1).

3.4.1.3.2 Imported transient voltage durability. The regulator shall not be damaged by 200 V spikes by the alternator, or by positive or negative 250 V spikes imported into the regulator on the remaining leads. Subsequently, the regulator shall meet the requirements of 3.4.1.1, 3.4.1.2, and 3.4.1.5 (see 4.6.4.3.2).

3.4.1.3.3 Exported transient voltage. As the result of alternator load dumps, the regulator shall not export, on its power or enable leads, positive or negative voltage of magnitudes exceeding 250 V peak for a period exceeding 0.001 second (see 4.6.4.3.3).

3.4.1.3.4 Steady state voltage. With no electrical load (battery and resistive loads disconnected), the regulated voltage, including ripple, shall not rise above 40 V peak (see 4.6.4.3.4).

3.4.1.4 Electromagnetic interference (EMI). Electromagnetic emissions shall be limited as specified in 3.4.1.4.1 and 3.4.1.4.2 (see 4.6.4.4).

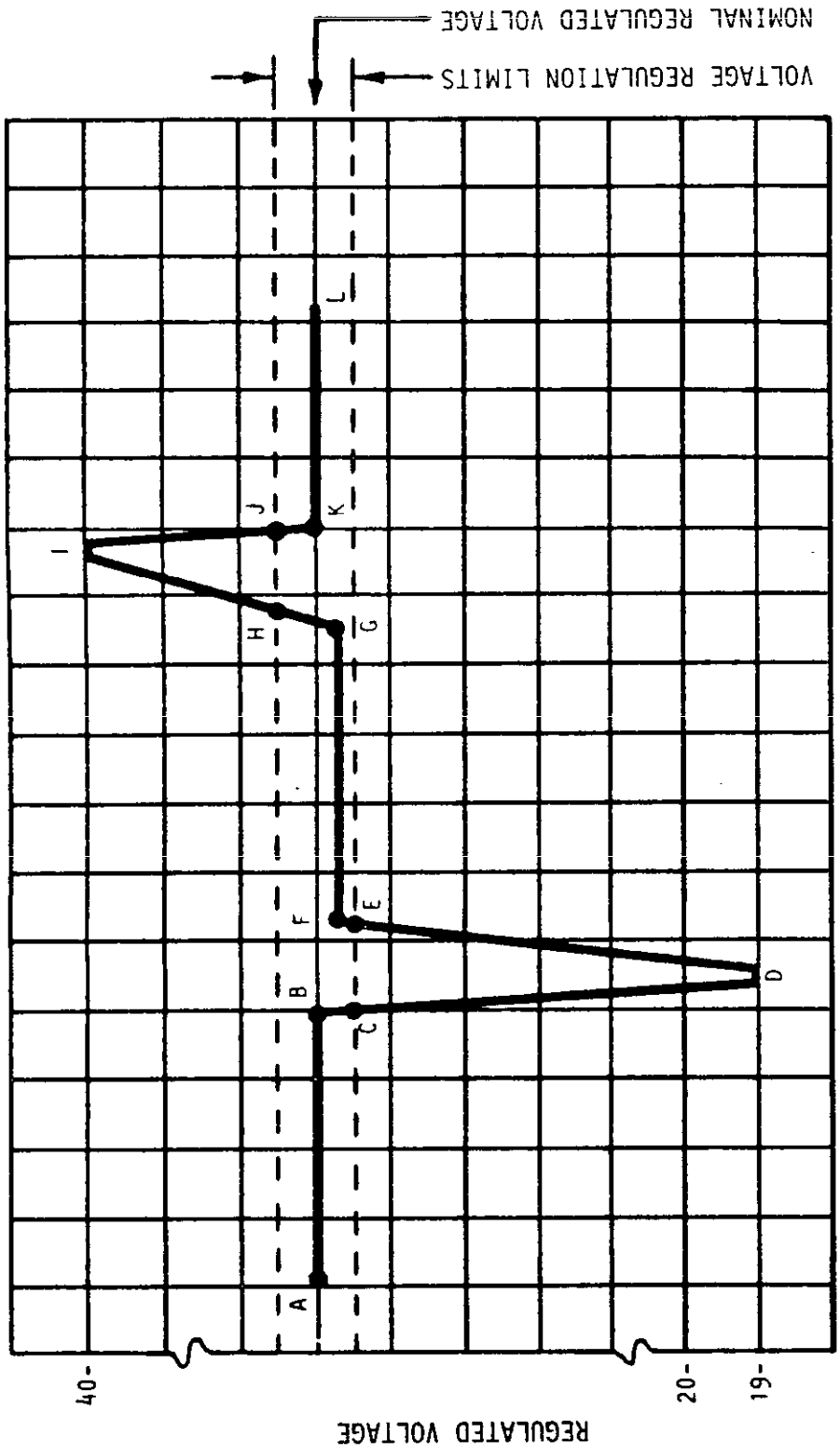
3.4.1.4.1 Conducted emissions. Conducted electromagnetic emissions in the frequency range of 10 kiloHertz (kHz) to 50 megaHertz (MHz) shall not appear on the battery, load, or control leads in excess of the values shown in figure 3.

3.4.1.4.2 Radiated emissions. Broadband E-field emissions in the specified frequency range shall not be generated and radiated in excess of the values shown in figure 4 at a test distance of three feet (1 meter). In the frequency range of 30 MHz to 10 gigahertz (GHz), the limit shall be met for both horizontally and vertically polarized waves.

3.4.1.5 Leakage current. When 25 $\pm$ 1 V is applied between the voltage regulator positive and negative battery terminals, the current flow shall be less than 0.1 milliamperes (ma) (see 4.6.4.5).

3.4.1.6 Field circuit resistance. Regulator field circuit resistance shall be not more than 0.2 ohms when the regulator is full on (see 4.6.4.6).

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:24Z
Check the source to verify that this is the current version before use.



(A-B) - BATTERY LOAD ONLY PRIOR TO LOAD APPLICATION
(B) - POINT OF LOAD APPLICATION
(C-E) - VOLTAGE UNDERSHOOT RECOVERY TIME
(D) - UNDERSHOOT VOLTAGE
(F-G) - REGULATED VOLTAGE (ER) DURING APPLIED LOAD (STEADY STATE)
(G) - POINT OF LOAD DISCONNECTION
(H-J) - VOLTAGE OVERSHOOT RECOVERY TIME
(I) - OVERSHOOT VOLTAGE
(K-L) - BATTERY VOLTAGE AFTER LOAD DISCONNECT
(C-E) - MAX. TIME LIMIT - 0.125 SEC
(H-J) - MAX. TIME LIMIT - 0.300 SEC

FIGURE 2. Voltage excursion limits.

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:24Z
Check the source to verify that this is the current version before use.

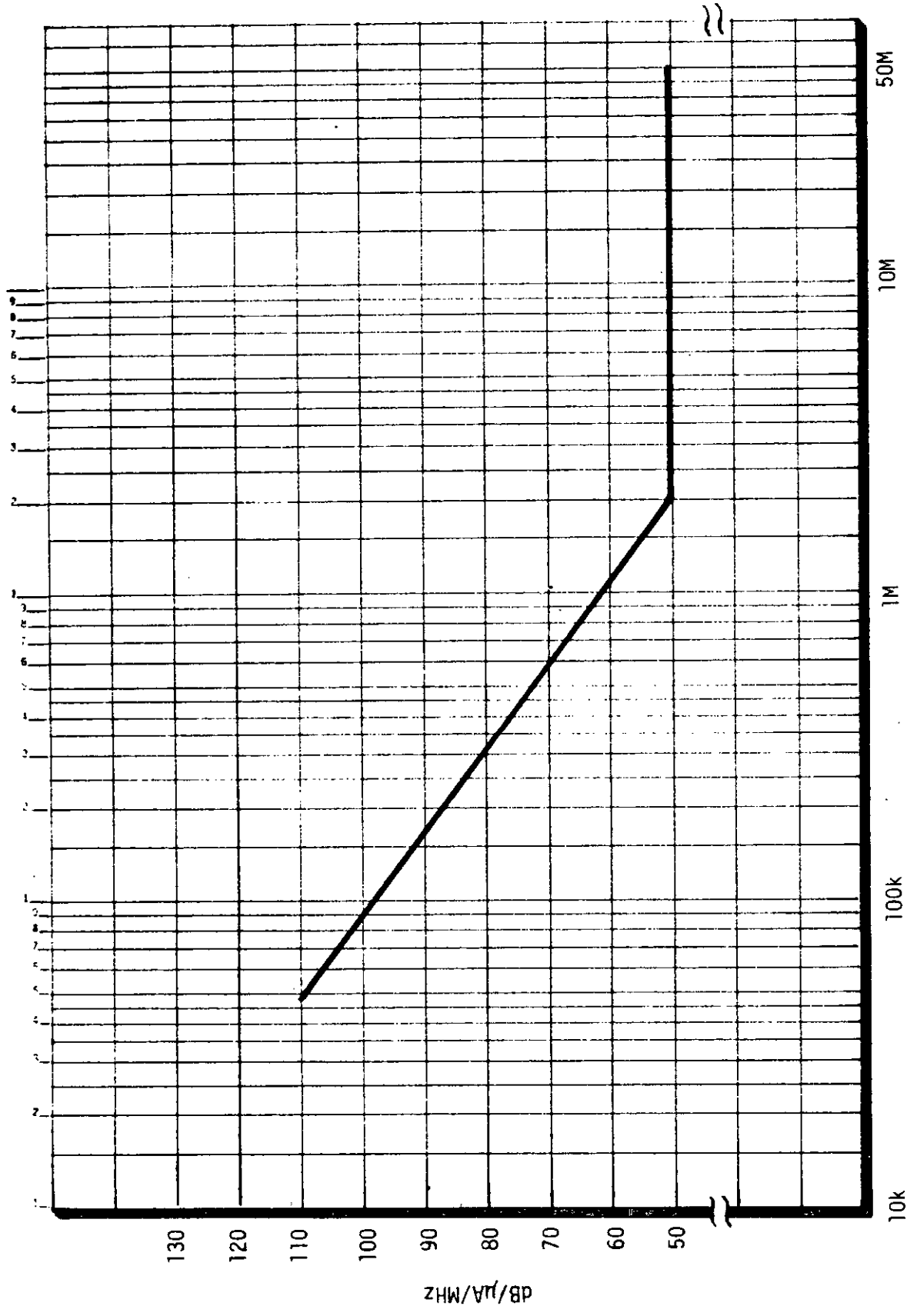


FIGURE 3. Broadband emissions.

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:24Z
Check the source to verify that this is the current version before use.

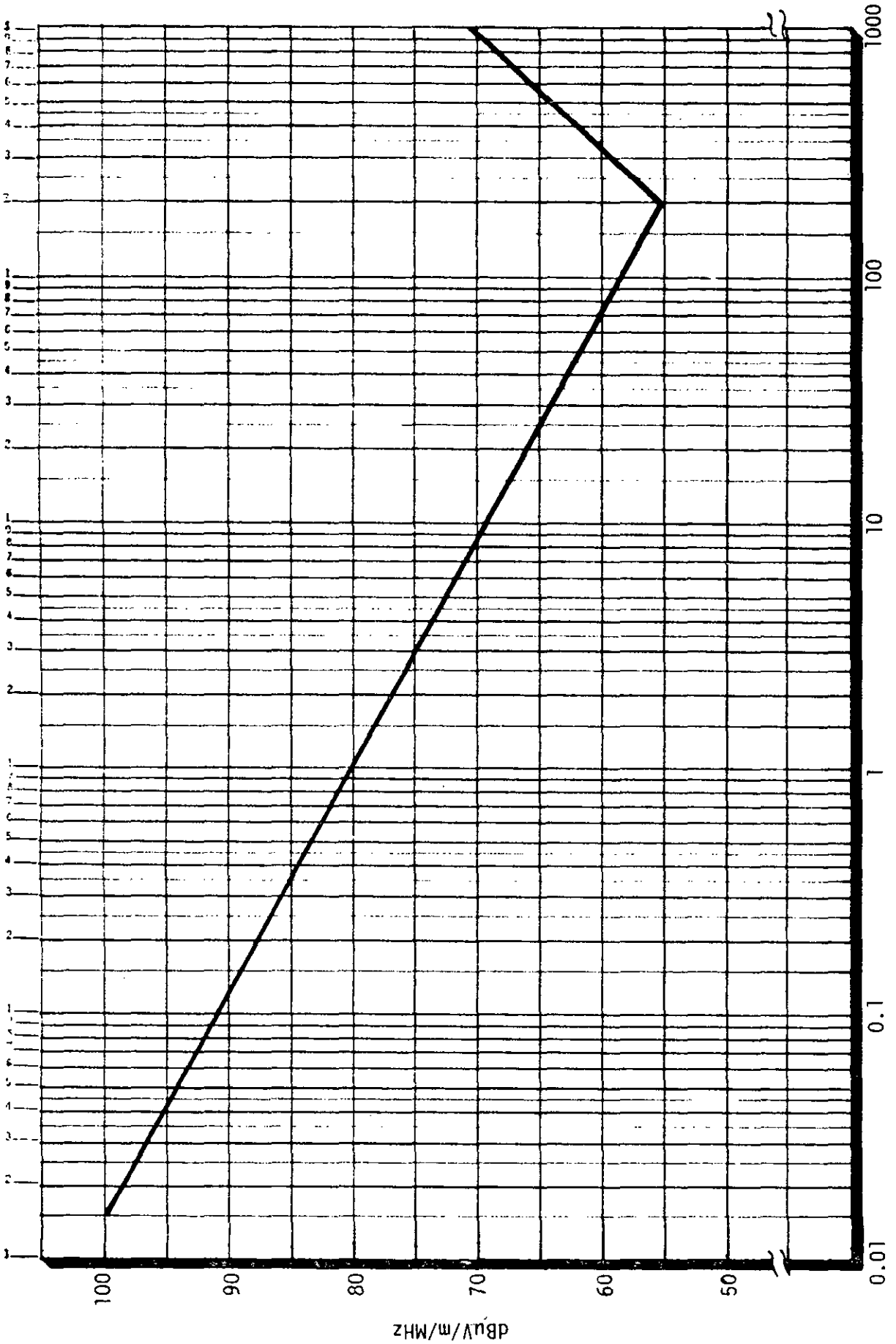


FIGURE 4. Broadband emissions.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:24Z
Check the source to verify that this is the current version before use.

3.4.1.7 Endurance. The regulator shall be capable of withstanding the endurance load cycles specified in table I for a total of 1000 hours. The regulated voltage shall not deviate from the voltage stability requirements of 3.4.1.1. There shall be no servicing, adjustments, or replacement of parts throughout the test. Subsequently the regulator shall meet the requirements of 3.4.1.1 through 3.4.1.1.2 and 3.4.1.2, 3.4.1.5, 3.4.1.8, 3.4.1.9 and 3.4.2.6 (see 4.6.4.7).

TABLE I. Regulator endurance cycle.

Alternator speed (rpm)	Duration (Hours)	Regulator ambient air temperature °F (°C)
2500 ± 100	250	77 ± 15 (25 ± 8)
2500 ± 100	100	225 ± 5 (107 ± 3)
6000 ± 100	299 3/4	77 ± 15 (25 ± 8)
8000 ± 100	350	77 ± 15 (25 ± 8)
10 000 ± 200	1/4	77 ± 15 (25 ± 8)

3.4.1.8 Reverse polarity protection. With the application of reverse polarity, the voltage regulator shall protect the generating system from damage. The generating system shall subsequently meet the requirements of 3.4.1.1, 3.4.1.2 and 3.4.1.5 without any repairs or adjustments (see 4.6.4.8).

3.4.1.9 Overvoltage protection. The regulator shall incorporate a resettable overvoltage protection scheme to protect the remainder of the vehicle electrical system from excessive voltage if a fault develops which would cause the voltage of the generated power to rise above 32 V. Reaction time shall be between 0.25 and 1.00 second. Once the protection mode has been entered, the regulator shall not cycle from the protection mode to the operation mode, but shall stay in the protection mode (see 4.6.4.9).

3.4.1.10 Adjustment range. The regulated voltage shall be adjustable over the range of less than 26 to 31.5 ± 0.5 V dc. Activating overvoltage protection 3.4.1.9 shall constitute voltage regulator failure (see 4.6.4.10).

3.4.1.11 Enable system. The regulator shall be enabled (turned on) by applying electrical power between pins F and C of connector J-2 (see 4.6.4.11).

3.4.2 Environmental conditions.

3.4.2.1 Temperature range. The regulator shall not be damaged by storage at -80°F (-62°C). The nominal regulated voltage, between ambient temperatures of -65°F (-54°C) and 0°F (-18°C), shall be 29.5 V dc. Between ambient temperatures of 0°F (-18°C) and 160°F (71°C), the regulated voltage shall decrease linarly from 29.5 V at a rate of 0.015 V per °F (.027 V per °C) to 27.1 V dc. Above 160°F (71°C) and up to 225°F (107°C) ambient

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:24Z
Check the source to verify that this is the current version before use.

temperature, the regulated voltage shall be 27.1 V dc. The tolerance on the regulated voltage shall be ± 0.6 V. Figure 4 shows this temperature compensation curve. When the regulator is subjected to load changes in accordance with 3.4.1.1, regulated voltage shall remain within the limits thereby specified (see 4.6.5.1).

3.4.2.2 Fungus resistance. After exposure to a fungus-laden environment for 90 days, the regulator shall perform as specified in 3.4.1.1, 3.4.1.2, 3.4.1.5 and 3.4.2.6. No servicing, adjustment, or replacement of parts shall be necessary. The regulator shall not be adversely affected by fungi under conditions favorable for their development (see 4.6.5.2).

3.4.2.3 Saline atmosphere (corrosion) resistance. After exposure to a saline atmosphere for 200 hours, the regulator shall perform as specified in 3.4.1.1, 3.4.1.2, 3.4.1.5, and 3.4.2.6. No servicing, adjustment, or replacement of parts shall be necessary (see 4.6.5.3).

3.4.2.4 Shock resistance. After shock testing as specified in 4.6.5.4, the regulator shall evidence no breakage, loosening of parts, abrasion, or other physical damage. The regulator shall meet the performance requirements of 3.4.1.1, 3.4.1.2, 3.4.1.5 and 3.4.2.6 (see 4.6.5.4).

3.4.2.5 Vibration resistance. During vibration testing as specified in 4.6.5.5, the regulator shall meet the requirements of 3.4.1.1. The test time for each primary axis shall be at least six hours for a total of at least 18 hours. After testing as specified, the regulator shall evidence no breakage, loosening of parts, abrasion, or other physical damage. Subsequent to the test, the regulator shall meet the requirements of 3.4.1.1, 3.4.1.2, 3.4.1.5 and 3.4.2.6 (see 4.6.5.5).

3.4.2.6 Waterproofness. The regulator shall meet the waterproofness requirements specified for type II, class 1 components in MIL-STD-1184. Subsequent to exposure, the regulator shall perform as specified in 3.4.1.1, 3.4.1.2, and 3.4.1.5 (see 4.6.5.6).

3.5 Identification marking. The regulator shall be identified and marked in accordance with MIL-STD-130 (see 4.5.2 and 4.6.2).

3.5.1 Nameplate. A nameplate conforming to MIL-P-514 shall be attached to the regulator housing as specified on Drawing 10947439 (see 4.5.2 and 4.6.2). The legend on the plate shall include following information:

Regulator
28 volts dc
Manufacturer's identification
Serial number
Military part number
Federal stock number
Contract number
Date of manufacture
US

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:24Z
Check the source to verify that this is the current version before use.

3.6 Workmanship. Workmanship shall be of such quality as to insure that regulators furnished under this specification are, in appearance and actuality, free of defects that compromise, limit, or reduce performance in intended use. Regulators shall be free of burrs, chips, cracked insulation, loose connections or solder, rust, scratches, and sharp edges (see 4.5.2 and 4.6.2).

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order (see 6.2), the contractor is responsible for the performance of all inspection requirements as specified herein. The contractor is responsible for maintaining records of the specified inspections. The records shall be maintained for a minimum of three years after the physical completion of the contract. The records shall include, but not be limited to, numerical results for each required test for each item tested, date of tests, name(s) of personnel performing tests, equipment used, and equipment calibration records. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform or witness any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Inspection equipment. Unless otherwise specified in the contract (see 6.2), the supplier is responsible for the provision and maintenance of all inspection equipment necessary to assure that supplies and services conform to contract requirements. Commercial, modified commercial, or supplier designed inspection equipment or measuring set-ups must be capable of repetitive measurements to an accuracy of 10 percent of the measurement tolerance. Calibration of inspection equipment shall be in accordance with MIL-STD-45662.

4.2 Classification of inspection:

a. First article inspection (see 4.4).
b. Quality conformance inspections (see 4.5).
   1. Examination (see 4.5.2).
   2. Acceptance tests (see 4.5.3).
   3. Control tests (see 4.5.4).

4.3 Inspection conditions. Unless otherwise specified herein, the test conditions shall be as specified in 4.3.1, 4.3.2 and 4.3.3.

4.3.1 Test circuit. The test circuit shall consist of alternator 10947517, two MS35000 batteries (connected in series), resistive loads to simulate the remainder of the vehicle electrical system, and necessary connecting cables. The regulator connectors shall be mated with their respective mating connectors. The mating connector grommets shall have any unused holes sealed. The resistive load and battery shall be connected to pins B and A of connector J-2, respectively. Pin C of connector J-2 shall be

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:24Z
Check the source to verify that this is the currentversion before use.

electrically connected to the regulator housing with 4 gage wire. The regulator shall be enabled (turned on) by having the battery voltage applied to pin F of connector J-2. While operating, the regulator shall have the regulated voltage applied to pin F of connector J-2. The battery shall be in good condition, taking a charge current of not more than 4.5 amperes at the regulator set point.

4.3.2 Ambient air temperature. Testing shall be performed at 77 ± 15°F (25 ± 8°C) ambient air temperature.

4.3.3 Mounting for tests. Mounting for testing the regulator shall provide at least 1/4 inch (in) [6 millimeters (mm)] clearance between the regulator and the mounting surface during high temperature and endurance tests (see 4.6.4.7, 4.6.4.3.2, 4.6.2.4.6 and 4.6.4.9.4). During shock and vibration testing, the regulator shall be mounted as it is in a vehicle using only the four 10-32 holes shown on the drawing.

4.4 First article inspection. Unless otherwise specified (see 6.2), the Government shall select four regulators (see 6.2) for first article inspection. First article samples shall be examined as specified in 4.5.2, and numbered and tested in the sequence specified in table II. Approval of the first article sample by the Government shall not relieve the contractor of his obligation to supply regulators that are fully representative of those inspected as a first article sample. Any changes or deviation of the production units from the first article sample shall be subject to the approval of the contracting officer.

4.4.1 First article inspection failure. Failure of any sample to conform to any requirement specified herein shall be cause for the Government to withhold approval of first article inspection until action by the contractor to correct defects and prevent recurrence has been approved by the Government.

TABLE II. Classification of inspections.

Title	Requirement	Inspection	First article Sample No. 1	2	3	4	Quality conformance Acceptance (100%)	Control
Materials	3.2	4.6.1	X					
Defects	3.3, 3.5, 3.6	4.6.2	X					
Voltage adjustment	3.3.1	4.6.3	X					
Dissimilar metals	3.3.2	4.6.1	X					
Standard parts	3.3.3	4.6.1	X					
Threaded parts	3.3.4	4.6.1	X					
Solder	3.3.4		X					
Polarity	3.3.6	4.6.1	X					
Electromagnetic interference suppression	3.3.7	4.6.1	X					

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:24Z
Check the source to verify that this is the current version before use.

TABLE II. Classification of inspections - Continued.

Title	Requirement	Inspection	First article Sample No. 1	2	3	4	Quality conformance Acceptance (100%)	Control
Finish	3.3.8	4.6.1	X					
Crimp terminations	3.3.9	4.6.1	X					
Fuse or circuit breaker	3.3.10	4.6.1	X					
Overvoltage protection circuit	3.3.11	4.6.1	X					
Voltage regulation	3.4.1.1	4.6.4.1.1	X	X	X	X		X
Voltage regulation	3.4.1.1	4.6.4.1.2					X	
Low voltage	3.4.1.2	4.6.4.2	X	X	X	X		X
Ripple voltage	3.4.1.3.1	4.6.4.3.1	X	X	X	X		X
Imported transient voltage durability	3.4.1.3.2	4.6.4.3.2	X	X	X	X		X
Exported transient voltage	3.4.1.3.3	4.6.4.3.3	X	X	X	X		X
Steady state voltage	3.4.1.3.4	4.6.4.3.4	X	X	X	X		X
Electromagnetic interference	3.4.1.4	4.6.4.4				X		
Leakage current	3.4.1.5	4.6.4.5	X	X	X	X	X	
Field circuit resistance	3.4.1.6	4.6.4.6	X	X	X			
Endurance	3.4.1.7	4.6.4.7.1	X	X	X	X		
Endurance	3.4.1.7	4.6.4.7.2						X
Reverse polarity protection	3.4.1.8	4.6.4.8		X	X		X	
Overvoltage protection	3.4.1.9	4.6.4.9	X	X	X	X		X
Adjustment range	3.4.1.10	4.6.4.10		X	X		X	
Enable system	3.4.1.11	4.6.4.11	X		X			X
Temperature range	3.4.2.1	4.6.5.1		X	X			X
Fungus resistance	3.4.2.2	4.6.5.2			X			X
Saline atmosphere resistance	3.4.2.3	4.6.5.3		X				X
Shock resistance	3.4.2.4	4.6.5.4		X	X			
Vibration resistance	3.4.2.5	4.6.5.5		X	X			
Waterproofness	3.4.2.6	4.6.5.6.1		X	X			X
Waterproofness	3.4.2.6	4.6.5.6.2					X	

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:24Z
Check the source to verify that this is the current version before use.

4.5 Quality conformance inspection.

4.5.1 Lot formation. A inspection lot shall consist of all the regulator assemblies from an identifiable production period, from one manufacturer, submitted at one time for acceptance.

4.5.2 Quality conformance examination.

4.5.2.1 Sampling for examination. Samples for quality conformance examination shall be selected in accordance with general inspection level II of MIL-STD-105.

4.5.2.2 Acceptable quality level. Each regulator selected in accordance with 4.5.2.1 shall be examined to determine conformance to the following acceptable quality levels (AQL), on the basis of percent defective:

Classification	AQL
Major	1.0
Minor	2.5

4.5.2.3 Classification of defects. For examination purposes, defects shall be classified as specified in table III.

TABLE III. Classification of defects.

Categories	Defects	Method of examination
Critical	None	
Major	AQL 1.0% Defective	
101	Dimensions affecting interchangeability, out of tolerance (see 3.3)	SIE 1/
102	Faulty workmanship affecting function (see 3.6).	Visual
103	Marking, improper (see 3.5).	Visual
Minor	AQL 2.5% Defective	
201	Dimensions not affecting interchangeability, out of tolerance (see 3.3)	SIE 1/
202	Finish, improper (see 3.3.8)	Visual
203	Faulty workmanship affecting appearance (see 3.6).	Visual

1/ SIE = Standard Inspection Equipment.

4.5.3 Quality conformance tests.

4.5.3.1 Acceptance tests (100 percent). Each regulator shall be tested according to and in the sequence of table II.

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:24Z
Check the source to verify that this is the current version before use.

4.5.4 Control tests.

4.5.4.1 Sampling. Samples for control testing shall be selected at the rate of three from each 500 units produced, except that not less than three nor more than six units shall be selected from each month's production. The control test samples shall be selected from a lot which has passed the examination specified in 4.5.2 and the acceptance tests listed in table II. The control test samples shall be tested according to and in the sequence listed in table II.

4.5.5 Test failure. Failure of a test sample to pass quality conformance inspection shall be cause for Government to refusal to accept regulators until corrective action taken by the contractor has been approved by the Government.

4.6 Methods of inspection.

4.6.1 Materials and construction. To determine conformance to 3.2 and 3.3.2 through 3.3.11, inspection and material certification records shall be maintained by the contractor. Records shall be subject to review by the Government and shall include date, part, or characteristic identification, inspection results, and disposition of lot (accepted or rejected). Corrective action taken on noted defects shall be subject to approval by the Government.

4.6.2 Defects. Conformance to 3.3, 3.5 and 3.6 shall be determined by examination for the defects listed in table III. Examination shall be visual, tactile, or by measurement with standard inspection equipment.

4.6.3 Voltage adjustment. Conformance to 3.3.1 shall be determined by examination of the voltage regulator and drawings and design data.

4.6.4 Electrical.

4.6.4.1 Voltage regulation. To determine conformance to 3.4.1.1, tests specified in 4.6.4.1.1 or 4.6.4.1.2 shall be performed as applicable.

4.6.4.1.1 Voltage regulation - first article or control test. A regulator set according to 4.6.4.1.2, shall be operated at the set point current and alternator speed until it is temperature stabilized. Regulated voltage shall be at the set point value (see 6.5.4). Subsequently, the regulator shall be subjected to the tests specified in table IV. When tested as specified in table IV the resistive load shall be changed in less than 0.03 seconds to cause the load current to change from battery only to the specified value, and to return to battery only.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:24Z
Check the source to verify that this is the current version before use.

TABLE IV. Voltage regulator tests.

Alternator speed - rpm	Load current - A
2000 ± 100	Battery only
"	50 ± 2
"	95 ± 5
6000 ± 100	Battery only
"	50 ± 2
"	95 ± 5
8000 ± 100	Battery only
"	50 ± 2
"	95 ± 5

4.6.4.1.2 Voltage regulation - acceptance test. The regulator shall be adjusted to operate, when it is temperature stabilized, at its set point.

4.6.4.2 Low voltage test. To determine conformance to 3.4.1.2, the regulator test circuit shall be as specified (see 4.3.1), except a diode of adequate current capacity shall be connected in the circuit between pin A of connector J-2 and the battery positive terminal. The diode cathode shall be connected to the battery terminal. Then the alternator shall be operated at 2000 ± 100 rpm, the resistive load adjusted to a convenient value, and no more than 16 V dc applied between pins F and C, and A and C of connector J-2. Then the resistance of the resistive load shall be lowered until the alternator operates in its current limiting mode (at least 95 A load current) and the regulated voltage drops to less than 16 V dc. The generating system shall continue to deliver the load current (at least 95 A) with the regulated voltage between 14 and 16 V for a minimum of ten minutes. Subsequently the resistance of the load shall be raised, and the regulator operated at 25 ± 5 A until the temperature is stabilized. The regulated voltage shall be as specified in 3.4.1.1.

4.6.4.3 Transient and steady state voltage characteristics. To determine conformance to 3.4.1.3, the regulator shall be tested in accordance with 4.6.4.3.1 through 4.6.4.3.4. Transient, ripple, and peak voltages shall be measured using a cathode ray oscilliscope.

4.6.4.3.1 Ripple voltage. To determine conformance to 3.4.1.3.1, the regulator shall be tested as specified in 4.6.4.1.1. The voltage between pin pairs A and C, and B and C of connector J-2 shall be measured.

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:24Z
Check the source to verify that this is the currentversion before use.

4.6.4.3.2 Imported transient voltage durability. To determine conformance to 3.4.1.3.2, the regulator shall be tested as specified in 4.6.4.3.2.1 and 4.6.4.3.2.2. During these tests the regulator shall be enabled by connecting in series with pin F of connector J-2 a 500 ± 50 ohm resistor, a 450 ± 50 microHenry inductor, and a separate power supply. The voltage between pins F and C of connector J-2 shall be adjusted to 24 ± 1 V. A battery resistance load or alternator shall not be included in this test circuit. The regulator shall be temperature soaked for 12 hours at the specified temperatures before the start of applicable tests. Subsequent to these tests the regulator shall be tested, without any adjustment or repair, in accordance with 4.6.4.1, 4.6.4.2 and 4.6.4.5.

4.6.4.3.2.1 Imported from alternator. A capacitor of at least 75 microFarads, charged to at least a voltage of 200 V dc, shall be discharged between sockets C and D of connector J-1. The charged capacitance shall be applied once every six seconds, equally spaced, for at least 8 hours. The 8 hours shall be divided between 2 hour periods at -65 ± 5°F (-54 ± 3°C) and 6 hour periods at 225 ± 5°F (107 ± 3°C) ambient temperatures.

4.6.4.3.2.2 Imported from system. A capacitor of at least 0.48 microfarad charged to a voltage of at least 250 V dc shall be discharged separately between each of the pin pairs A and C, B and C, and F and C of connector J-2. The charged capacitances shall be applied once every 2 seconds, equally spaced, for a period of two hours. The test shall then be repeated with the capacitor charging voltage reversed. The test shall be performed at a temperature of 225 ± 5°F (107 ± 3°C).

4.6.4.3.3 Exported transient voltage. To determine conformance to 3.4.1.3.3, the alternator shall be operated at 6000 ± 200 rpm and the regulator load current shall be set to at least 100 A. The regulator shall be enabled as in 4.6.4.3.2. The load current shall be interrupted in less than 0.03 seconds, to generate an exported voltage. Voltage between the pin pairs A and C, B and C, and F and C shall be separately monitored for exported transient voltages from before to at least one second after the load interruption. This procedure shall be repeated 50 times. The exported transient voltage is the peak voltage measured during the test.

4.6.4.3.4 Steady state voltage. To determine conformance to 3.4.1.3.4, the alternator shall be operated at 6000 ± 200 rpm and the regulator load current set to at least 60 A. The load current shall be interrupted in less than 0.03 seconds. The voltage between each pin pair A and C, and B and C of connector J-2 shall be separately monitored for at least 5 minutes after interruption of the load current. The exported transient voltage measured in 4.6.4.3.3 shall not be counted in determining the voltage rise.

4.6.4.4 Electromagnetic interference. Conformance to 3.4.1.4 shall be determined by tests in accordance with 4.6.4.4.1, 4.6.4.4.2, and the appendix to this specification. The alternator and regulator shall be operated at the respective speeds and loads specified in table IV.

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:24Z
Check the source to verify that this is the current version before use.

4.6.4.4.1 Conducted emissions. Conducted emissions levels shall be determined by testing in accordance with Method 1 (see 60). When authorized by the procuring activity because of present high ambient signals, this test may be supplemented with tests in accordance with Method 2 (see 70) and Method 3 (see 80).

4.6.4.4.2 Radiated emissions. Radiated emission levels shall be determined by testing in accordance with Method 4 (see 90).

4.6.4.5 Leakage current. To determine conformance to 3.4.1.5, the resistive load shall be disconnected and no voltage shall be applied to pin F of connector J-2. The current draw of the regulator from the battery shall be measured.

4.6.4.6 Field circuit resistance. To determine conformance to 3.4.1.6, the generating system shall be operated to cause the regulator to operate in its "full on" mode with the load current at least 95 A. Resistance shall be computed by summing the voltages measured between sockets C and B, and sockets E and D of connector J-1 and dividing the sum by the alternator current (the current flowing from socket B of connector J-1 to the alternator field). The resulting value is in ohms.

4.6.4.7 Endurance. To determine conformance to 3.4.1.7, the regulator shall be operated as specified in 4.6.4.7.1 or 4.6.4.7.2 according to the test classification. The oven used for the 225°F (107°C) portion of the specified tests shall use no mechanical means of stirring or circulating the air. All air movement shall be by natural convection. Temperature measurement in the oven shall be made one to two inches above the top center of the regulator.

4.6.4.7.1 Endurance-first article test. The regulator shall be subjected to the endurance cycle specified in table I. The duration of the test shall be further divided into one hour load cycles, consisting of at least 15 minute intervals operating at the following four load current levels:

a. Battery only
b. 10 $\pm$ 2 A
c. 50 $\pm$ 2 A
d. At least 98 A

During testing the specified load currents shall be obtained by adding to or removing from the resistive load used for the previous load level resistance. The load change shall be accomplished in less than 0.03 seconds. The regulated voltage shall remain as specified (see 3.4.1.1). After the regulator has been subjected to each load cycle for five to ten minutes, the regulator shall be de-energized (turned off) and re-energized (turned on), once each at load leavels a, b, and d; and twice at c. The off time shall be 1 to 2 seconds. When the regulator is de-energized, the battery shall not be required to power the resistive load. This one hour load cycle shall be

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:24Z
Check the source to verify that this is the currentversion before use.

repeated throughout the 1000 hour test duration. Prior to beginning the 225°F (107°C) portion of the endurance test, the regulator shall be conditioned at the test ambient temperature for at least four hours. At the 10 000 rpm alternator speed condition, the electrical load shall be continuously varied from battery only to 98 A. Subsequently, the regulator shall meet the requirements of 4.6.1.1, 4.6.4.2, 4.6.4.5, 4.6.4.8, 4.6.4.9 and 4.6.5.5 without any repairs or adjustments.

4.6.4.7.2 Endurance - control test. The voltage regulator shall be subjected to the 225°F (107°C) portion of the endurance cycle specified in table I. It shall be continuously enabled and the load current shall be 90 ± 5 A. Subsequently, the regulator shall meet the requirements of 4.6.4.1, 4.6.4.2, 4.6.4.5, 4.6.4.8, 4.6.4.9, and 4.6.5.6 without any repairs or adjustments.

4.6.4.8 Reverse polarity protection. To determine conformance to 3.4.1.8, the regulator shall be disconnected from the battery and resistive load. With the alternator stationary, minus 28 ± 1 V shall be applied between pins A, B, and F in parallel, and pin C of connector J-2 for ten seconds. Subsequently, the regulator shall meet the requirements of 4.6.4.1, 4.6.4.2, and 4.6.4.5 without any repairs or adjustments to it or the specified alternator.

4.6.4.9 Overvoltage protection. To determine conformance to 3.4.1.9, the regulator shall be tested according to 4.6.4.9.1 through 4.6.4.9.4 for first article and control tests, or according to 4.6.4.9.5 for acceptance tests. The test circuit shall be as specified above, except that a diode shall be connected between pin A of connector J-2 and the battery positive terminal. The anode of the diode shall be connected to the positive terminal of the battery. Any need to replace fuses or other regulator parts during over voltage tests, except only as allowed in 4.6.4.9.3 shall constitute a failure.

4.6.4.9.1 Low limit. A voltage of 31.95 ± 0.05 V shall be applied for at least ten minutes between pin pairs A (cathode terminal of the diode) and C, B and C, and F and C of connector J-2. During this time, the voltage between sockets C and D of connector J-1 shall not differ from the applied voltage by more than 0.5 V.

4.6.4.9.2 Failure due to nuisance tripping. The regulator shall not require replacement as the result of overvoltage conditions that do not result from a regulator failure causing it to go "full-on". The regulator shall be subject to the tests specified in 4.6.4.9.2.1 and 4.6.4.9.2.2.

4.6.4.9.2.1 Operation due to transient overvoltages. The applied voltage (see 4.6.3.9.1) shall be removed, the resistive load adjusted to draw 77 ± 2 A, the alternator set to a speed of at least 6000 rpm and the regulated voltage measured. An overvoltage of 33 ± 1 V shall be applied and maintained for at least five seconds between pin pairs A and C, and B and C of connector J-2. Reaction time shall be measured between the application of overvoltage and the drop of the voltage between socket pairs C and D, and B and D of connector J-1 to less than 15 V. Subsequent to the application of the overvoltage not more than 0.5 A shall flow to the regulator through pins F, A or B of connector J-2.

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:24Z
Check the source to verify that this is the current version before use.

4.6.4.9.2.2 Manual reset. After the elapse of the specified five seconds (see 4.6.3.9.2), the resistive load shall be disconnected, and power from pin F of connector J-2 and the specified overvoltage shall be removed for a period of 10 to 15 seconds. Power shall be reapplied to pin F of connector J-2 and the resistive load reconnected. Regulated voltage shall be within 0.2 V of the regulated voltage measured in 4.6.4.9.2.1. The tests specified in 4.6.4.9.2.1 and in this paragraph shall be repeated for four additional trials.

4.6.4.9.3 Operation due to "full-on" regulator failure. The alternator speed and resistive load shall be as specified in 4.6.4.9.2.1. A fault, causing the regulator to go "full-on", shall be simulated by shorting across the output terminals of the alternator field current final output stage. The terminals shorted shall be recorded. Reaction time is the time between the rise of alternator voltage (between sockets C and D of connector J-1) above 32 V and the drop of the voltages between pin pairs A and C, and B and C of connector J-2 below 30 V. The voltage between sockets C and D of connector J-1 shall not rise over 200 V peak. The voltage between socket pairs C and D, and B and D of connector J-1 shall drop below 15 V in less than five seconds after the rise of the alternator voltage above 32 V. Within 0.03 seconds after the voltage between pins B and C of connector J-2 has dropped below 30 V, the resistive load shall be disconnected. The voltage at the diode cathode shall be above 21 V. The current flow through any of the pins A, B, or F of connector J-2 shall not be greater than 0.5 A. Without any adjustments or changes in the specified test circuit, the current and voltages shall remain as specified herein for at least ten minutes after the operation of the overvoltage feature. If a circuit breaker or fuse is used in the regulator it may be reset or replaced. Otherwise no regulator parts shall be replaced, changed or adjusted.

4.6.4.9.4 Extreme temperature operation. Before testing, the regulator shall be conditioned for at least 12 hours at the test temperature. Subsequent to temperature conditioning, the regulator shall be tested as specified in 4.6.4.9.1, 4.6.4.9.2, 4.6.4.9.3, and 4.6.4.10 at each of the specified test temperatures. The test temperatures shall be minus 65 $\pm$ 5°F (-54 $\pm$ 3°C) and 225 $\pm$ 5°F (107 $\pm$ 3°C).

4.6.4.9.5 Overvoltage protection - acceptance testing. In lieu of the circuit specified by 4.3.1, the contractor may use his test circuit, which shall consist of the necessary equipment and cables to perform the specified tests. The regulator shall have 31.95 $\pm$ 0.05 V dc applied between sockets C and D of connector J-1, and pins F and $\overline{C}$ of connector J-2. The voltage between pin pair A and B, and pin C of connector J-2, shall be at least 31 V. The voltages shall be as specified herein for a minimum of five seconds. The applied voltage shall be raised to 34.05 $\pm$ 0.05 V. Within the specified time limits (see 3.4.1.9), the voltages at pin pairs A and B of connector J-2 shall drop below 24 V.

4.6.4.10 Adjustment range test. To determine conformance to 3.4.1.10, the regulator shall be adjusted to operate at its set point. The regulated voltage shall then be adjusted by rotating the voltage adjustment rheostat to its limits of rotation.

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:24Z
Check the source to verify that this is the currentversion before use.

4.6.4.11 Enable system test. To determine conformance to 3.4.1.11, the regulator shall have battery voltage applied between pins F and C of connector J-2. The regulator shall be operated at its set point. Power shall be removed from pin F of connector J-2. The voltage between pins A and C of connector J-2 shall be 24 ± 1 V.

4.6.5 Environmental conditions.

4.6.5.1 Temperature range. To determine conformance to 3.4.2.1, a regulator meeting the requirements of 3.4.1.1 shall be soaked for at least 24 hours at minus 80 ± 5°F (-62°C ± 3°C). Ambient air temperature shall be measured one to two inches above the top center of the regulator. Subsequently, the regulator shall be cold soaked for another period of at least 24 hours at minus 65 ± 5°F (-54 ± 3°C). It shall then be operated at its set point in this latter environment until the temperature is stabilized. Subsequently, it shall be tested at alternator speeds and load currents specified in 4.6.4.1.1. The regulated voltage for the measured ambient air temperature shall be within the specified limits. This procedure, except the -80°F (-62°C) soak, shall be repeated for air temperatures of 0 ± 5°F (-18 ± 3°C), 160 ± 5°F (71 ± 3°C) and 225 ± 5°F (108 ± 3°C).

4.6.5.2 Fungus resistance test. To determine conformance to 3.4.2.2, the regulator shall be subjected to the test procedures specified in MIL-F-13927, for class 1 components, method B. During fungus testing, the regulator shall not be electrically operated, be part of an electrical circuit, or be otherwise disturbed. The regulator shall subsequently be subjected to tests specified in 4.6.4.1, 4.6.4.2, 4.6.4.5 and 4.6.5.5 without any repairs or adjustments.

4.6.5.3 Salt fog. To determine conformance to 3.4.2.3, the regulator shall be subjected for at least 200 hours to test procedures specified in MIL-STD-202, method 101. During testing, the regulator shall not be operated and its connectors shall be mated with the mating connectors, attached to cables necessary to perform subsequent tests. The regulator shall subsequently, without removal of the mating connectors, pass the tests specified in 4.6.4.1.1, 4.6.4.2, 4.6.4.5, and 4.6.5.5 without any repairs or adjustments.

4.6.5.4 Shock resistance. To determine conformance to 3.4.2.4, the regulator alone shall be subjected to test procedures specified in MIL-STD-202, method 213, condition I. Following shock testing, the regulator shall be subjected to the tests specified in 4.6.4.1, 4.6.4.2, 4.6.4.5, and 4.6.5.5 without repairs or adjustments.

4.6.5.5 Vibration resistance. To determine conformance to 3.4.2.5, the regulator shall be subjected to the test procedure specified in MIL-STD-202, method 204, condition B, frequency range 10 to 500 Hz (see 4.3.3). Connecting cables shall have an unsupported length of at least eight inches (20.3 cm) between its supports and the cable connectors. While being subjected to the specified vibration testing, the regulated voltage shall be measured and the regulator shall be tested according to 4.6.5.5.1 or 4.6.5.5.2 as applicable. The Government representative shall inspect the

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:24Z
Check the source to verify that this is the current version before use.

extent necessary to verify the applicable test method. Subsequent to vibration testing, the voltage regulator shall be subjected to the tests specified in 4.6.4.1, 4.6.4.2, and 4.6.4.5 without repairs or adjustment.

4.6.5.5.1 Regulator not equipped with fuse or circuit breaker. The regulator shall be subjected to the 77 ± 15°F (25 ± 8°C) portion of the endurance test (see 4.6.4.7.1).

4.6.5.5.2 Regulator equipped with a fuse or circuit breaker. The regulator shall be subjected to the 225 ± 5°F (107 ± 3°C) portion of the endurance test (see 4.6.4.7.1).

4.6.5.6 Waterproofness. To determine conformance to 3.4.2.6, the regulator shall be subjected to tests specified in 4.6.5.6.1 or 4.6.5.6.2, according to test classification.

4.6.5.6.1 Waterproofness - first article. The regulator shall be subjected to test procedures specified in MIL-STD-1184, method 100, procedure 1. Subsequent to waterproofness tests, the regulator shall be subjected to performance tests 4.6.4.1, 4.6.4.2, and 4.6.4.5.

4.6.5.6.2 Waterproofness - control and acceptance testing. The regulator shall be subjected to test procedures specified in MIL-STD-1184, method 100, procedure 5. The method of applying specified internal pressure is optional. The regulator shall be dry before packaging.

5. PACKAGING

5.1 Preservation, packaging, packing, and marking. Preservation, packaging, packing, and marking for the desired level of protection shall be in accordance with the applicable packaging standard or packaging data sheet specified by the contracting authority (see 6.2).

6. NOTES

6.1 Intended use. The regulator covered by this specification is intended for use in the charging circuit of the M113 personnel carrier and in other tactical or combat military vehicles under extreme conditions of climate and terrain.

6.2 Ordering data. Acquisition documents should specify the following:

a. Title, number, and date of this specification.
b. Drawing number of voltage regulator required (see 2.1.2).
c. If first article samples are not required (see 3.1).
d. If responsibility for inspection shall be other than as specified (see 4.1).
e. If responsibility for inspection equipments shall be other than as specified (see 4.1.1).
f. If inspection conditions shall be other than as specified (see 4.3).
g. If first article inspection is not required (see 4.4).
h. Applicable levels of preservation, packaging, packing, and marking (see 5.1).

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:24Z
Check the source to verify that this is the current version before use.

6.3 Recycled materials. The use of recycled materials which meet the requirements of the applicable material specifications without jeopardizing the intended use of the item shall be encouraged (see 3.2).

6.4 Definitions. For the purpose of this specification, the following definitions are used.

6.4.1 Temperature stabilized regulator. A regulator is temperature stabilized when, operated under constant conditions, the temperature of the center of an exterior surface has not changed by more than 1°F (0.5°C) in 15 minutes.

6.4.2 Connectors. Connector J-1 is part number 7064706 on Drawing 10947439. Connector J-2 is part number 7064429 on Drawing 10947939.

6.4.3 Load current. Load current is the current delivered by the alternator through the regulator to the batteries and resistive load.

6.4.4 Regulated voltage. Regulated voltage is the voltage output of the regulator measured between pins A and C of connector J-2.

6.4.5 Voltage convention. When a voltage (positive), is described as being measured between two points of a circuit, the first point is assumed to be positive with respect to the second. When a voltage measurement is described as negative, the first point is assumed to be negative with respect to the second.

6.4.6 Set point. Regulator set point is a regulated voltage deviating from the nominal voltage (see 3.4.2.1) by less than $\pm$ 0.2 V dc, at 25 $\pm$ 2 amperes load current, at an alternator speed of 2000 $\pm$ 100 rpm at regulator ambient air temperatures of 75 $\pm$ 30°F (24 $\pm$ 17°C).

6.5 Metric units. The use of symbols, and the conversion and rounding of values expressed in the International System of units are in accordance with ASTM-E-380.

6.6 Test methods. The test methods specified in the appendix were extracted from MIL-STD-462, Notice 3(EL), 9 February 1971 as follows:

Appendix	MIL-STD
Method 1	Method CE04
Method 2	Method CE02
Method 3	Method CRE02
Method 4	Method RE02

6.7 Criticality. This is a critcal item.

Custodian:
Army - AT

Preparing activity:
Army - AT

(Project No. 2920-A386)

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:24Z
Check the source to verify that this is the currentversion before use.

APPENDIX

TEST METHODS FOR ELECTROMAGNETIC INTERFERENCE,
VOLTAGE REGULATOR, SOLID STATE

10. Scope. This appendix covers test methods for determining conformance to requirements and levels of electromagnetic interference (EMI) emissions (conducted and radiated) for automotive regulators. This appendix is a mandatory part of the specification. The information contained herein is intended for compliance.

20. APPLICABLE DOCUMENTS

20.1 Government documents.

20.1.1 Specifications, standards and handbooks.

EL-CD-6003-0009A	- Electromagnetic Interference Measurement Instrumentation, Characteristics for.
MIL-STD-220	- Method of Insertion Loss Measurement.
MIL-STD-285	- Attenuation Measurements for Enclosures, Electromagnetic Shielding, for Electronic Test Purposes, Methods of.
MIL-STD-831	- Test Reports, Preparation of.

20.1.2 Drawings and publications.

ES-DL-198697	- Line Impedance Stabilization Network 2 MHz to 65 MHz.
SK.N 1791	- Line Impedance Stabilization Network 10 kHz to 10 MHz.
ES-F-201286	- Antenna, Biconical 30 MHz to 300 MHz, less Balun.
ES-DL-176439	- Bifalar Balun, 30 MHz to 300 MHz.
62J4040	- Antenna, Conical Log Spiral, 200 to 1000 MHz.
62J4041	- Antenna, Conical Log Spiral, 1 to 10 GHz.
DL-ES-A-217563	- Antenna, Double Ridged Horn, 200 to 2000 MHz.
DL-ES-A-217311	- Antenna, Double Ridged Horn, 1 to 12.4 GHz.

Copies of these drawings may be obtained from the contracting officer of from the following activities:

62J4040 and 62J4041
Hq, USAF Logistics Command
Engineering Data Center (SGCDC)
Wright Patterson Air Force Base, Ohio 45433

US Naval Applied Science Laboratory
Brooklyn, New York, N.Y.

Others: SK.N 1791

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:24Z
Check the source to verify that this is the currentversion before use.

APPENDIX

Commanding General,
US Army Electronics Command
ATTN: AMSEL-TD-TE-C
Fort Monmouth, New Jersey 07703

30. Test conditions.

30.1 Ambient electromagnetic level. The radient ambient electromagnetic field level, during testing, measured with the test sample de-energized shall be at least 6 decibel (db) below the allowable specified limit. Precautions for avoiding ambient signals generated by test equipment shall be observed. At the lower frequencies, where screen room filtering may be inadequate and where the power line 10 microFarad (uF) capacitor represents a significant impedance, various methods for verifying the source of an excessive emission can be utilized, such as the technique discussed in the notes to Method 2, and shall be made a subject of the test report. When measuring the ambient field level, all the equipment necessary to perform the specified testing, excepting the regulator subject of the test, shall be operated. The alternator shall be operated at the maximum speed (excepting overspeed) and electrical load specified by this specification. When the alternator is an integrally regulated unit (the regulator is directly attached, internally or externally, to the alternator), in lieu of operating the alternator as specified herein, the alternator drive may be operated at the speed necessary to attain the required alternator speed. If a test site does not meet the requirements specified herein, measurements shall be performed using the procedure specified in Method 3. When instrumentation for performing Method 3 is not available, the contractor may use an equivalent method with the approval of the procuring activity.

30.2 Ground plane. A copper or brass ground plane (solid plate) shall be used that has a minimum thickness of 0.25 mm for copper or 0.63 mm for brass and is 2.25 square meters or larger in area with the smaller side no less than 76 cm. The generator and ground plane shall be bonded to the shielded room such that the dc bonding resistance shall not exceed 2.5 milliohms. In addition, the bonds shall be placed at distances no greater than 90 cm apart. Unless specified otherwise by the basic specification, the testing shall be performed with the negative leads bonded and not bonded to the ground plane.

30.3 Accessory equipment precaution. Care shall be taken to insure that accessory equipment (spectrum monitors, oscilloscopes, earphones and other equipments) used in conjuction with interference meters do not affect measurement integrity.

30.4 Excess personnel and equipment. The shielded enclosure shall be kept free of unnecessary equipment, cable racks, and desks. Only the equipment essential to the test being performed shall be in the enclosure. Personnel not actively involved in the test shall not be permitted in the enclosure.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:24Z
Check the source to verify that this is the current version before use.

30.5 Radio frequency (rf) absorber material. Rf absorber material may be used in shielded enclosures to reduce reflections from the surfaces of the enclosure to the measurement antennas. It is encouraged that this material be used immediately behind the test specimen against the enclosure wall in line with the test antennas.

30.6 Test site - shielded enclosures. Shielded enclosures shall be of sufficient size to adquately accept the item under test without sacrificing test accuracy or requiring deviation from the methods specified herein. Shielding and filtering characteristics shall meet the following minimum requirements:

a. Shielding effectiveness to electric fields and plane waves, as measured in accordance with MIL-STD-285, shall be at least 80 dB over the frequency range of the test.

b. When ac power is brought into the enclosure, power line filtering must be included and shall have an attenuation to frequencies above 10 kHz of at least 80 db as measured in accordance with MIL-STD-220.

40. Measuring equipment. The following describes the test equipment used in the test methods contained in this appendix.

40.1 Test antennas. Table V lists antennas which shall be used for performing radiated emission (RE) measurements. Where antenna frequency ranges overlap either may be used, receiver sensitivity permitting. EL-CD-6003-0009A states requirements and testing techniques for antennas used in testing to this appendix; antenna factors shall be determined in accordance with the testing techniques of that specification.

40.2 Test antenna counterpoise (monopole). The test antenna counterpoise shall be referenced to the same ground reference used for the EMI meter. This requires bonding the counterpoise to the reference ground plane. The bond must be achieved by use of a solid metal sheet having the same width as the counterpoise and by welding along the entire edge at the points of contact. If desired the counterpoise may be configured so that one dimension is of adequate length to reach the equipment ground plane. The antenna matching network must be mounted below the counterpoise.

40.3 Other equipment characteristics. Requirements for measuring equipment characteristics are contained in EL-CD-6003-0009A along with techniques for measurement of these characteristics. Table VI represents a suggested list of equipments for each test. In some cases the list presents more than one model capable of performing a given function. Any equipment equivalent to those listed may be used with the approval of the procuring activity.

40.4 Use of measuring equipment. All laboratory equipment shall be operated as prescribed by the applicable instruction manuals unless otherwise specified herein. This appendix takes precedence in the event of conflict with instruction manuals or other such documents issued by industry or other Government agencies.

Source: https://assist.dla.mil -- Downloaded: 202640-26T04:24Z
Check the source to verify that this is the current version before use.

APPENDIX

TABLE V. Measurement equipment.

Antenna identification, nomenclature	Frequency range	Antenna factor (see figure)	Drawing number	Manufacturer
41" Rod Antenna and Matching Network	14 kHz to 30 MHz	Equipment Manual	None	Electro-Metrics; Singer-Metrics; Stoddart
Biconical Antenna w/ T-1 Balun	20 MHz to 200 MHz	Figure 5	ES-F-201286	EMCO; Honeywell
Biconical Antenna w/ Bifalar Balun	30 MHz to 300 MHz	Figure 6	ES-DL-176439	
Conical Logarithmic Spiral	200 to 1000 MHz	Figure 7	62J 4040	Stoddart; EMCO; Electro-Metrics
Conical Logarithmic Spiral	1 to 10 GHz	Figure 8	62J 4041	Stoddart; EMCO; Electro-Metrics
Linearly Polarized Double Ridged Waveguide Horn	200 to 2000 MHz	Figure 9	DL-ES-A 217563	EMCO
	1 to 12.4 GHz	Figure 10	DL-ES-A 217311	EMCO

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:24Z
Check the source to verify that this is the current version before use.

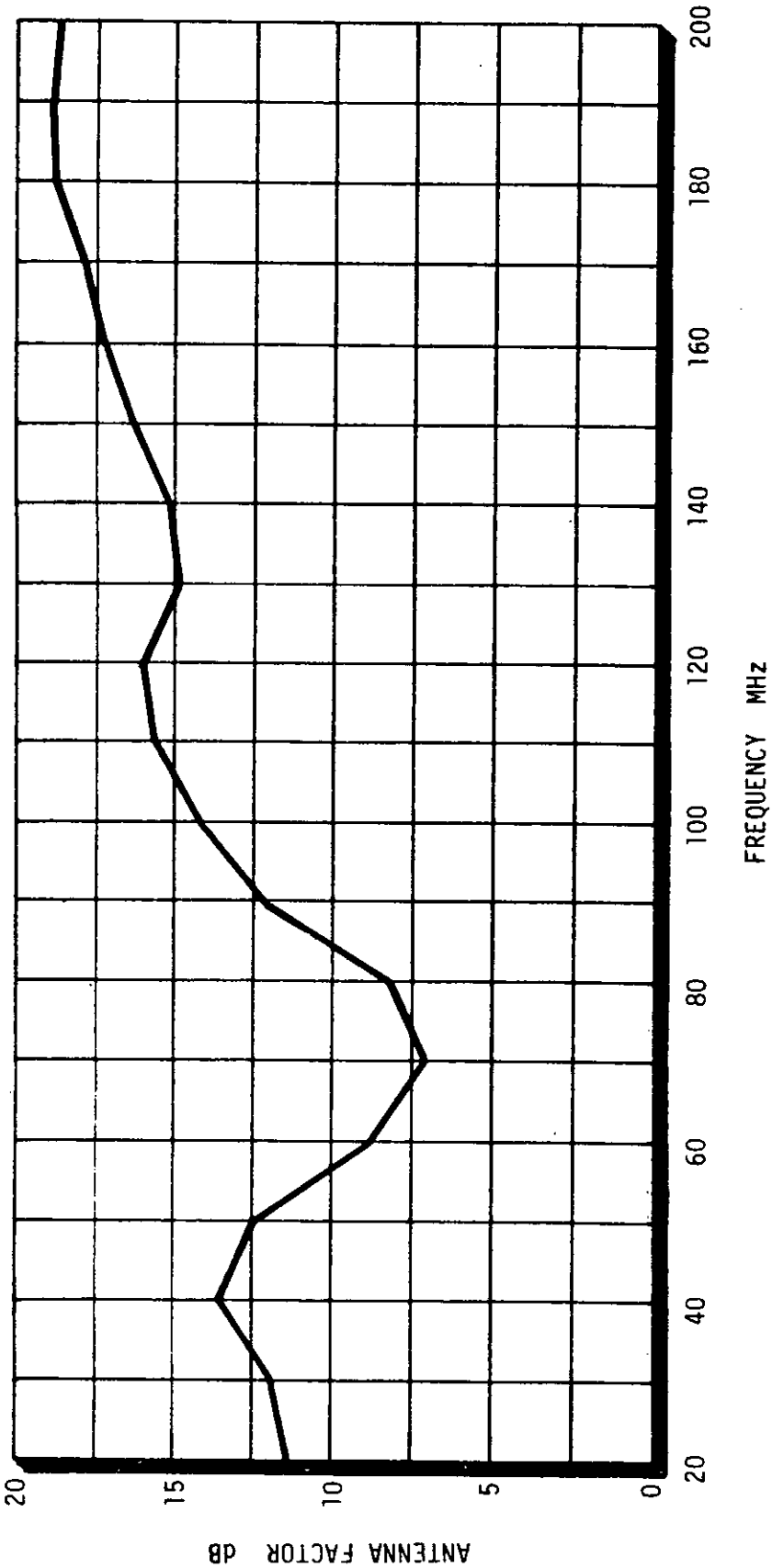


FIGURE 5. Antenna factor for biconical antenna.
(To be added to receiver meter reading in dBµV to convert to field intensity in dBµV/meter.)

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:24Z
Check the source to verify that this is the current version before use.

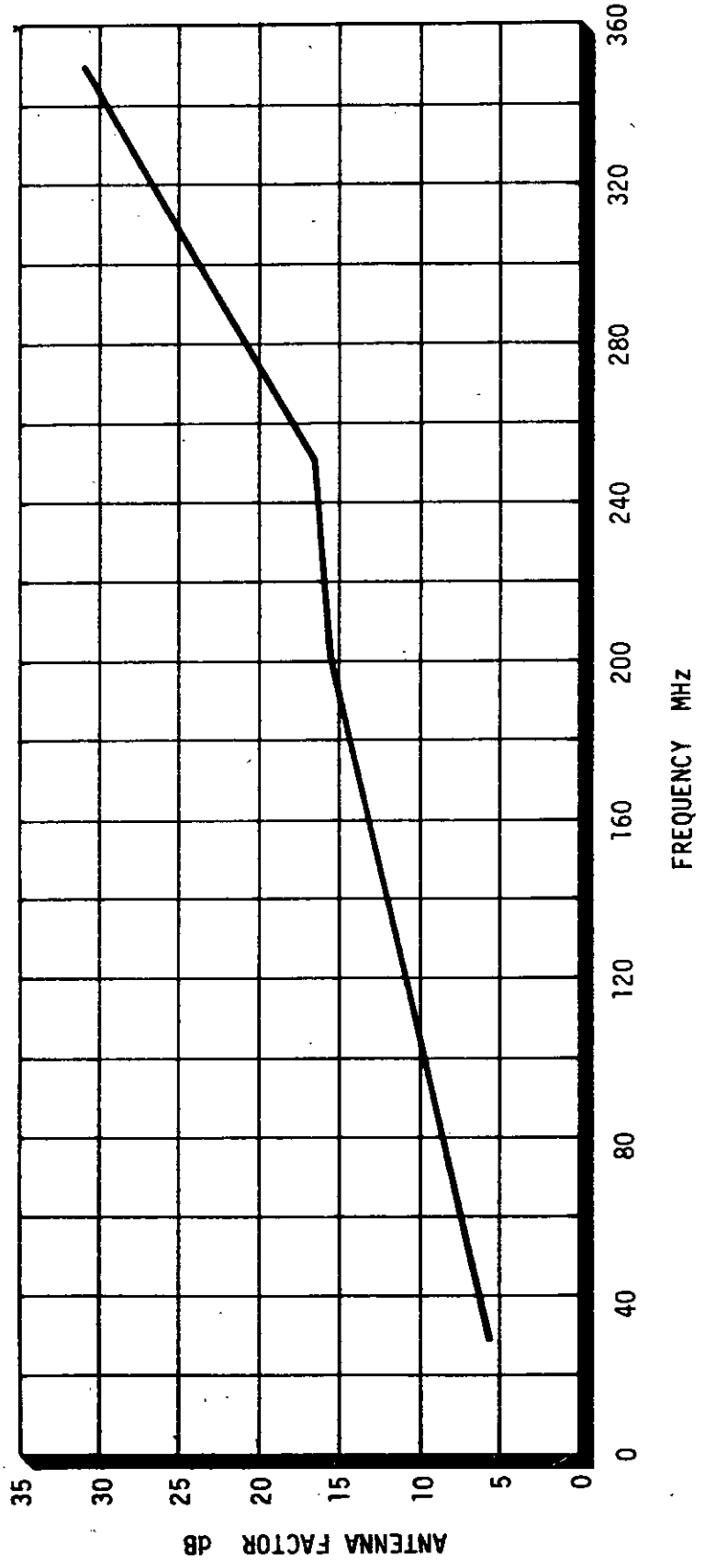


FIGURE 6. Antenna factor for biconical antenna with bifalar balun.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:24Z
Check the source to verify that this is the current version before use.

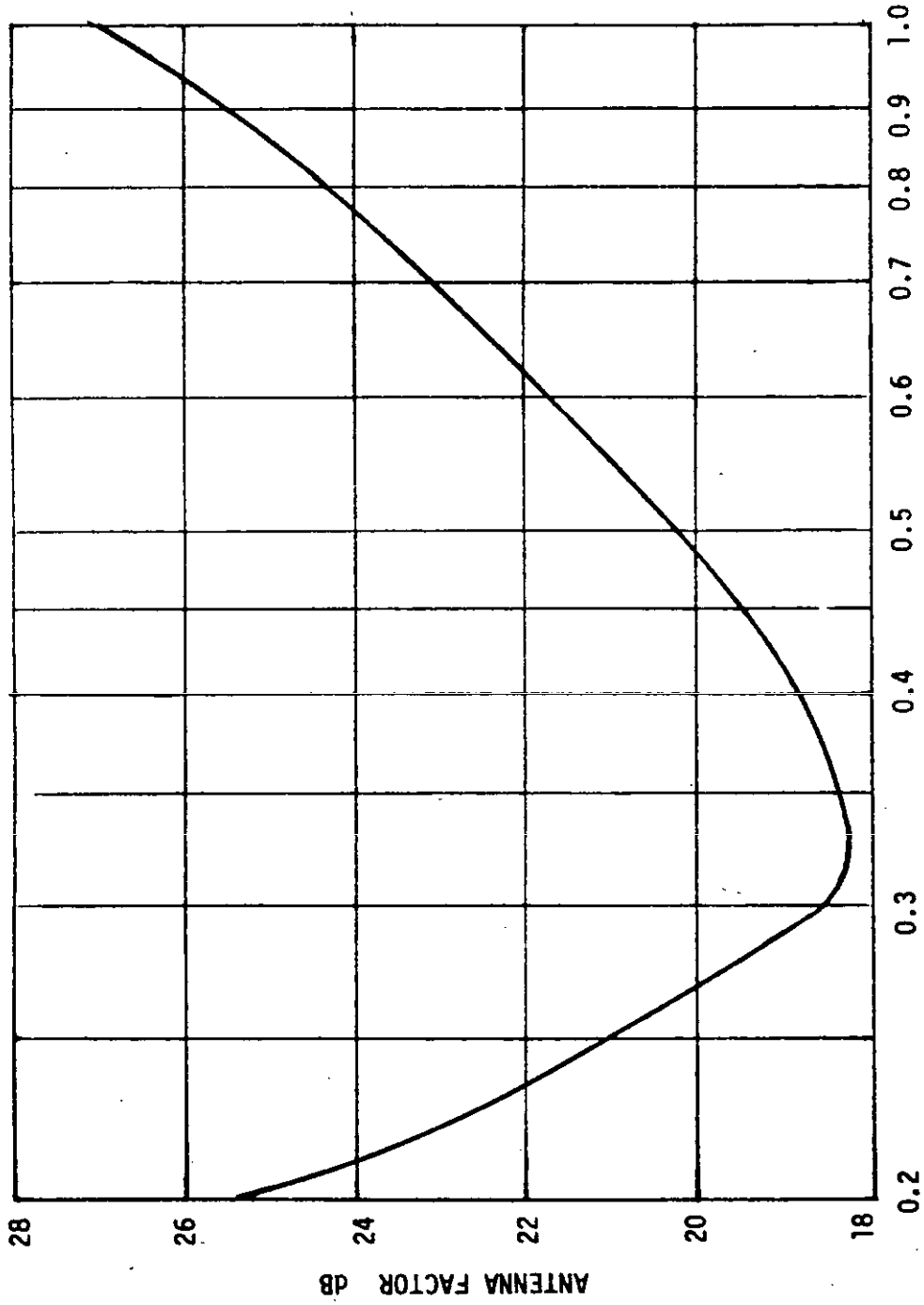


FIGURE 7. Antenna factor for log spiral.
(To be added to receiver reading in dBμV to convert to field intensity in dBμV/meter.)

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:24Z
Check the source to verify that this is the current version before use.

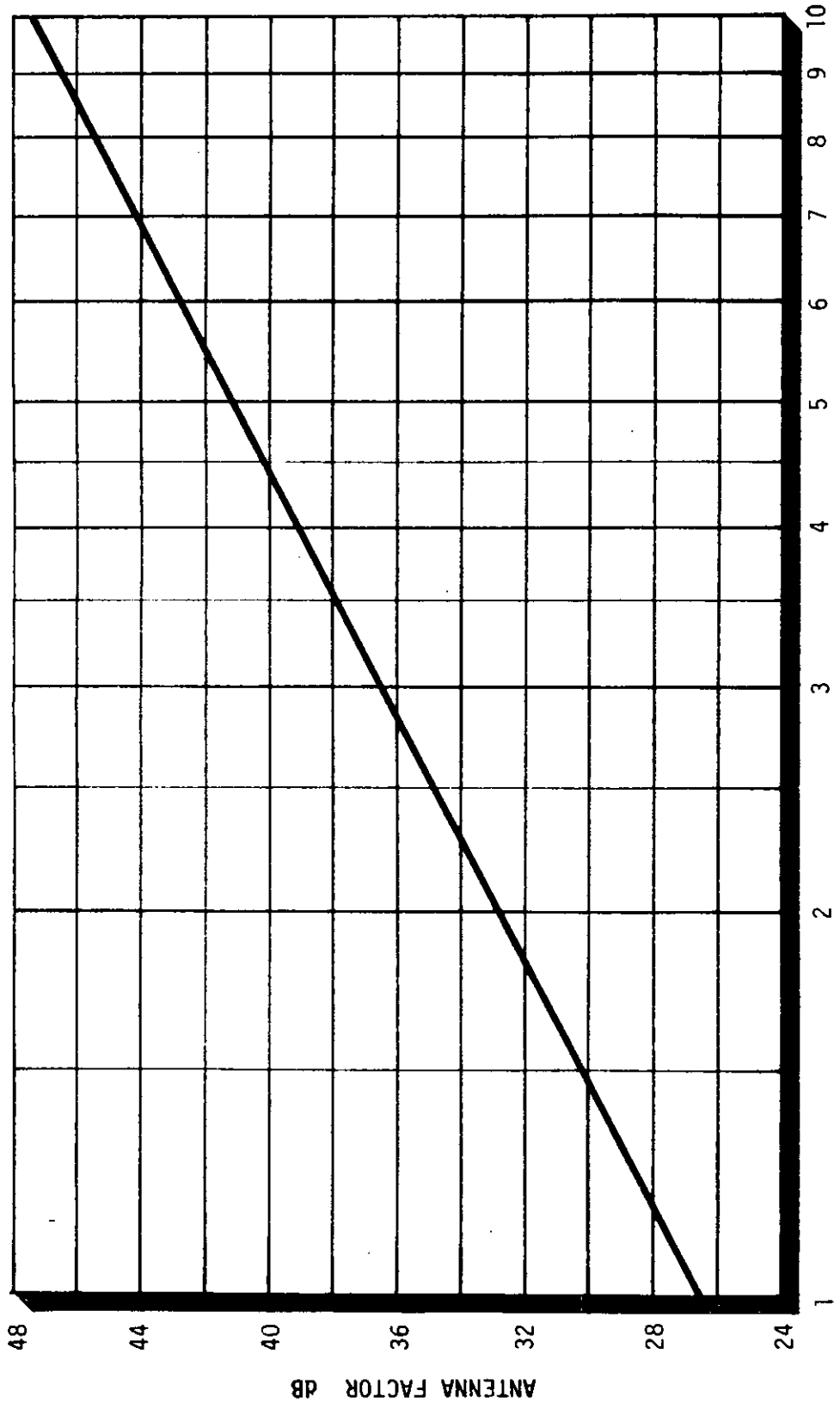


FIGURE 8. Antenna factor for log spiral.

(To be added to receiver meter reading in dBμV to convert to field intensity in dBμV/meter.)

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:24Z
Check the source to verify that this is the current version before use.

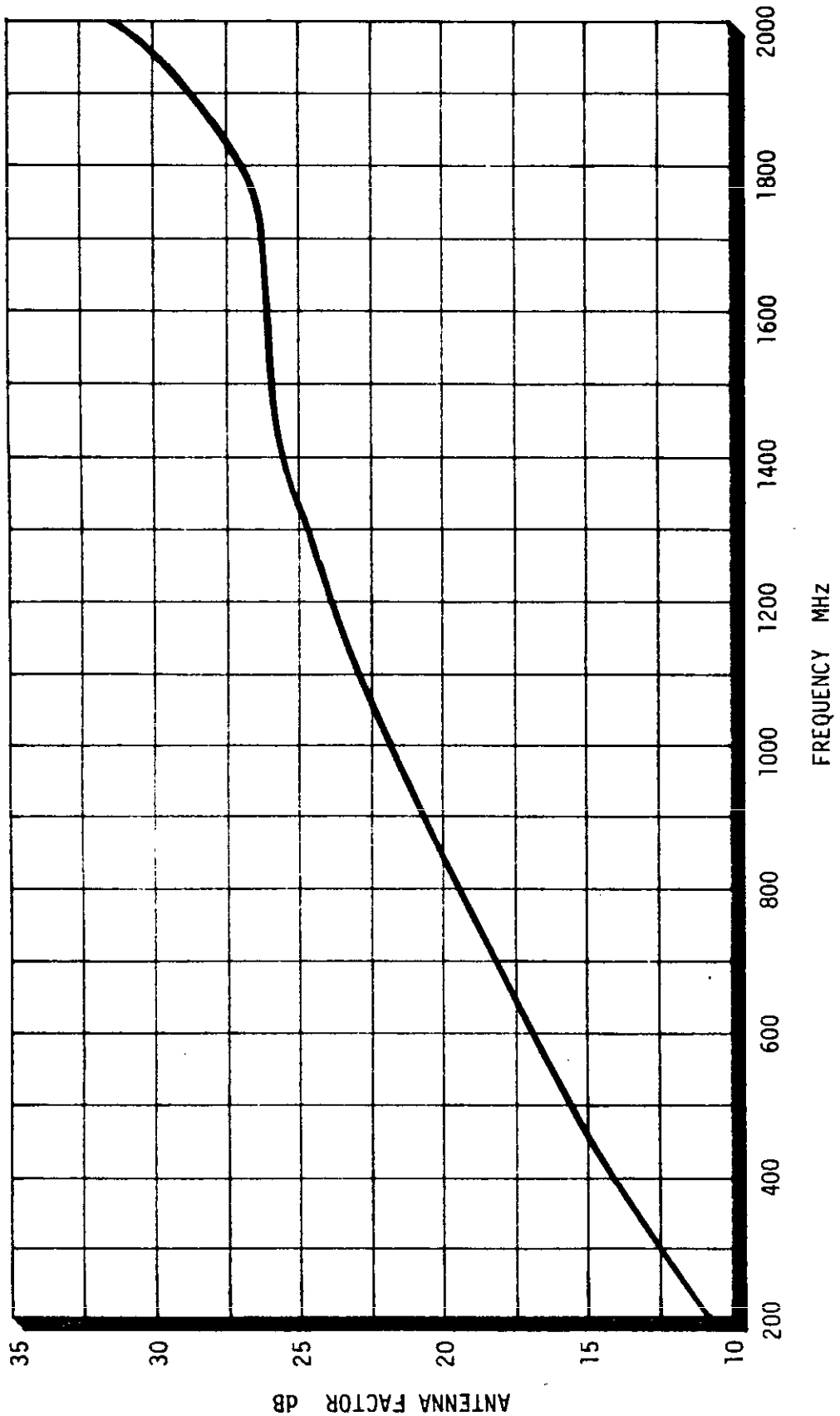


FIGURE 9. Antenna factor double ridge waveguide horn.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:24Z
Check the source to verify that this is the current version before use.

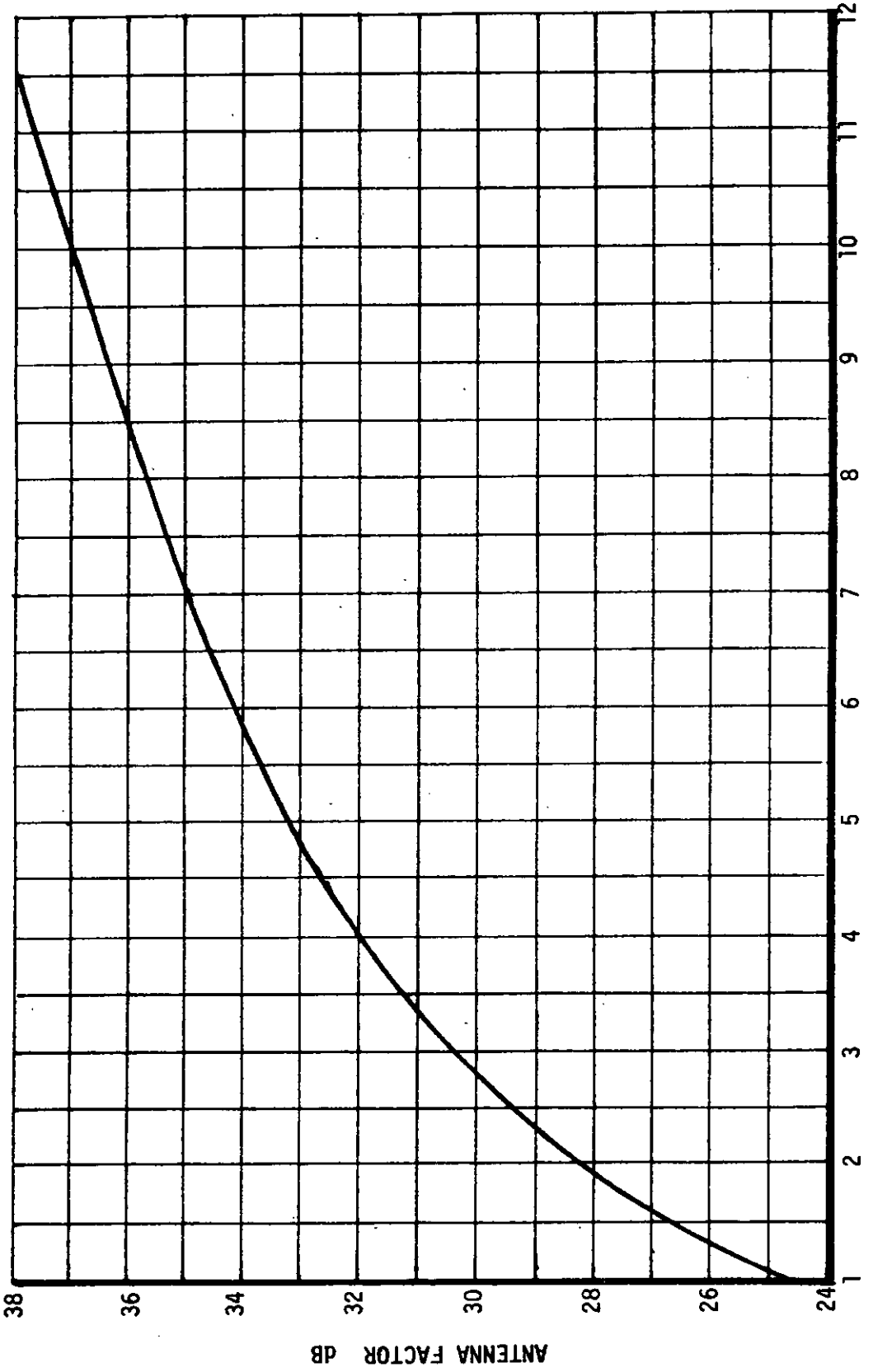


FIGURE 10. Antenna factor for ridged horn antenna.
(To be added to receiver meter reading in dBμV
to convert to field intensity in dBμV/meter.)

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:24Z
Check the source to verify that this is the current version before use.

APPENDIX

TABLE VI. Suggested test equipment or equivalent.

Equipment	Model	Frequency range	Test method 1	2	4
Current probes:					
Electro-Metrics	PCL-10	30 Hz to 50 kHz		X	
Electro-Metrics	PCL-10	30 Hz to 50 kHz	X		
Electro-Metrics	PCL-24	14 kHz to 100 MHz	X		
Stoddart	91550-1	1 MHz to 100 MHz	X		
Genistron	GLP-5120	10 Hz to 15 MHz		X	
Honeywell	3892	1 kHz to 5 MHz		X	
Audio frequency coupling transformer:					
Solar	6220-1A	20 Hz to 150 kHz		X	
Spectrum analyzer:					
Hewlett-Packard	8552A 8553L	1 kHz to 110 MHz	X	X	
Spectrum display units					
Electro-Metrics	SPD-123		X	X	X
Electro-Magnetics	Adapter		X		X
EMI meters:					
Singer-Metrics	NF-105A	14 kHz to 1 GHz	X		X
Singer-Metrics	EMA-910-10	1 to 22 GHz			X
Fairchild Electro-Metrics	EMC-10E	20 Hz to 50 kHz		X	
Fairchild Electro-Metrics	EMC-25R	14 kHz to 1 GHz	X		X
Stoddart NM62A	NM62A	1 to 10 GHz			X
Automatic scan system:					
Fairchild Electro-Metrics	FSS-250	14 kHz to 1 GHz	X		X
WEI	130A	20 Hz to 1 GHz	X	X	X
Line impedance stabilization networks:					
Filtron	FSQ-702S6	2 to 50 MHz		X	
Captor	LSN-112-115	2 to 50 MHz		X	
		10 kHz to 2 MHz		X	
Rejection networks:					
Electro-Metrics	HPK-10	400 Hz		X	
Electro-Metrics	NRG-10	400 Hz		X	

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:24Z
Check the source to verify that this is the current version before use.

40.4.1 Grounding of measuring equipment. Grounding of EMI instrumentation shall be accomplished in accordance with the following rules to avoid false data that may be introduced by ground loops.

a. The antenna shall be remote from the measuring instrument.
b. The EMI measuring instrument shall be physically grounded with only one connection to the equipment ground plane or to earth ground. The power line ground shall be broken by means of an isolation transformer.

40.4.2 Equipment warm-up time. Prior to commencing data collection, the measuring equipment shall have been switched on for a period of time adequate to allow parameter stabilization. If the operation manual does not specify a specific warm-up time, a period of one hour shall be observed.

40.4.3 Detector function. A peak detector shall be used for all measurements. Substitution generator output levels shall be corrected when the detector output and the signal generator output calibration are different functions of sine wave (peak, average, rms).

40.4.4 Monitoring of measuring equipment. The IF output of the EMI measuring instrument shall be monitored with a device that gives an amplitude versus frequency presentation on a cathode ray tube. This monitor is used to obtain information on the characteristics of the signals being measured. The measuring equipment shall also be monitored with headphones or a speaker.

40.4.5 Identification of spurious responses in measuring equipment. The measurement equipment shall be checked for spurious responses. False data caused by such spurious responses shall be so identified on the X-Y recordings or data sheets.

40.4.6 Calibration of measuring equipment. Measuring instruments and accessories used in determining compliance with this standard shall be calibrated under an approved program in accordance with MIL-C-45662. The calibration program document required by MIL-C-45662 shall be submitted for Government approval as part of the test plan. See EL-CD-6003-0009A for calibration methods for impulse generators.

40.4.7 Accuracy of measurement. All measurements made in accordance with this standard shall have the following accuracies. Proof of such accuracy shall be documented in the test report.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:24Z
Check the source to verify that this is the current version before use.

APPENDIX

a. Frequency accuracy:

Where specified limits are exceeded in the spurious response and harmonic and spurious emissions tests (antenna terminal), the frequency of measurement shall be accurate to within ± 0.1%. Since signal generators and EMI meters usually do not offer this accuracy, it will be necessary to employ a frequency counter or other similar standard. All other tests shall be subject to a frequency accuracy of ± 2%.

b. Amplitude accuracy:

Amplitude accuracies shall be ± 2 db.

40.5 Positioning of measurement antennas. Each face of the test sample shall be probed with a loop or other suitable sensor to determine the localized area producing maximum emission. Probing shall be performed at frequencies known or calculated to represent worst case interference; or, if no such information is available, probing shall be performed at no fewer points than one frequency for every two octaves over the frequency range of test. The face exhibiting worst case characteristics in any octave or band, provided that a band is not less than two octaves, shall face the test antenna for that portion of the frequency scan. Automatic scan techniques may be used to scan all sides.

a. When performing radiated emission tests no point of the antennas shall be less than 1 meter to the walls and 0.5 meters to the ceiling of the shielded enclosure or obstruction.

b. For radiated emission measurements between 30 MHz and 10 GHz linearly polarized antennas shall be positioned alternately to measure the vertical and horizontal components of the emission.

40.6 Measuring frequencies. The entire specified frequency range for each test shall be scanned. Measurements shall be taken at not less than three frequencies per octave representing the maximum indications within the octave.

40.7 Bonding of test sample. Except as specified herein, only the provisions included in the design of the equipment shall be used to bond it to the ground plane. When bonding straps are required to complete the test setup, they shall be identical to those specified in the installation drawings.

50. Test report.

50.1 Format. The general format of the test report shall be as specified in MIL-STD-831 with the modifications given herein.

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:24Z
Check the source to verify that this is the currentversion before use.

APPENDIX

50.1.1 Cover page. A cover page is required.

50.1.2 Administrative data. Data shall include but not be limited to the following:

a. Test performed by prime contractor or independent testing laboratory.
b. Contract number.
c. Authentication. Contractor personnel responsible for performance of the tests and witnessing organization or representatives.
d. Disposition of the test specimen.

50.1.3 Content. The test report shall contain but not be limited to the following:

a. Nomenclature of interference measuring equipment.
b. Serial numbers of interference measuring equipment.
c. Date of last calibration of interference measuring equipment.
d. Photographs or diagrams of test setup and test sample with identification.
e. Transfer impedance of current probes.
f. Antenna factors of specified antennas.
g. Measured level of emission at each frequency.
h. Graphs of applicable limits and measured data.
i. Antenna factors added to receiver reading (tabulation of receiver readings, antenna factors, frequency and antenna).
j. Measurement point.
k. Test method.
l. Data recorded.

1. Test frequency
2. Meter indication.
3. Attenuator setting.
4. Correction factors.
5. Corrected readings.
6. Ambient radiated electromagnetic levels.

50.1.4 Recommendations and conclusions. This section shall include the results of the tests in brief narrative form.

60. Method 1 - Conducted emissions 50 kHz to 50 MHz, power leads.

60.1 Applicability. This test method is applicable for measuring conducted emissions on dc power leads, including neutrals which are grounded externally to the equipment.

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:24Z
Check the source to verify that this is the current version before use.

60.2 Apparatus. Measurement instrumentation shall include:

a. Current probes.
b. Electromagnetic interference meter.
c. Headset and IF monitoring device.
d. 50 ohm resistive terminations.

60.3 Test procedure. The test setup shall be as shown in figure 11.

70. Method 2 - Conducted emission, 10 kHz to 50 kHz, ac power leads.

70.1 Applicability. This test method is applicable for measuring conducted emissions on ac power input and output leads, including neutrals which are grounded externally to the equipment and whose length exceeds 6 inches.

70.2 Apparatus.

a. Current probe.
b. Matching transformer, if required between the current probe and the interference meter to allow the measuring system to meet the sensitivity requirements needed to perform the test.
c. Electromagnetic interference meter.
d. Band, reject or high pass filter. A filter shall be used in the measuring circuit to attenuate all frequencies below 10 kHz.
e. Line impedance stabilization networks (LISN). See figures 12 and 13 for details of LISN covering applicable frequency range.
f. Headphones and intermediate frequency (IF) monitor.
g. 50 ohm resistive terminations for LISN's.

70.3 Test procedure. The test setup shall be as shown on figure 14. A meter bandwidth between 50 and 75 Hz shall be used.

Notes:

a. When matching transformers or band reject filters are used, their characteristics must be described in the test plan and test report.
b. Conducted emissions shall be measured separately on each power lead.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:24Z
Check the source to verify that this is the current version before use.

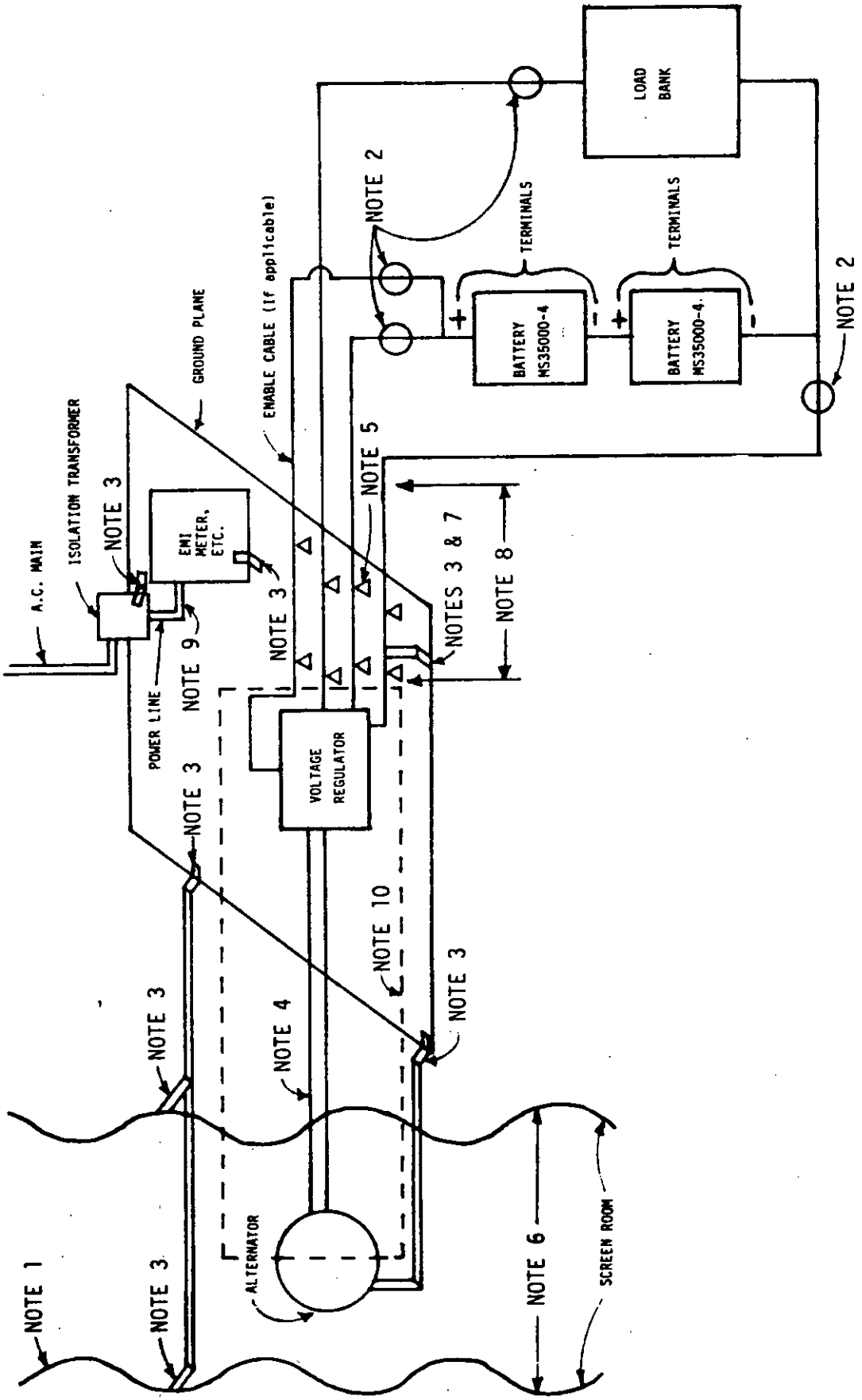


FIGURE 11. Test set-up for Method 1.

(See figure 11A for explanation of notes.)

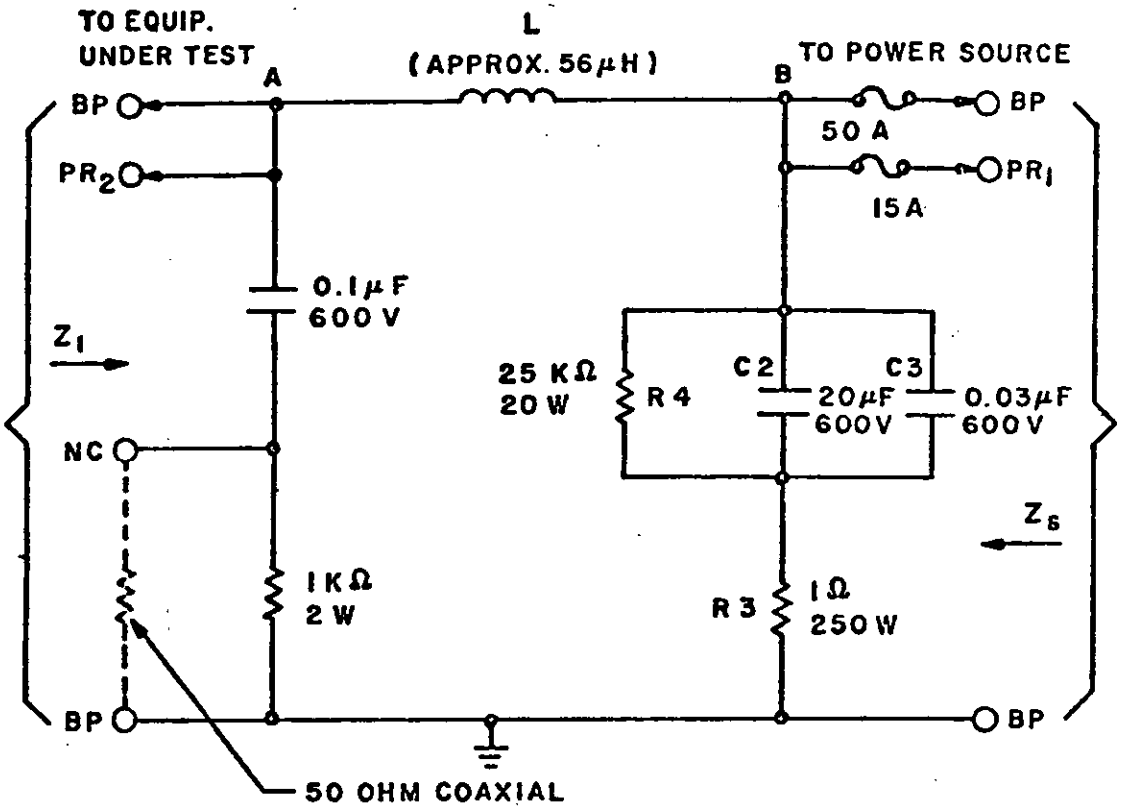
Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:24Z
Check the source to verify that this is the current version before use.

1. Set up for integrally regulated alternator.
2. Current probe.
3. Bonding strap.
4. Shielded dc output power and field cable. Use only with externally regulated alternators.
5. "△" is the symbol for stand offs that support the cables 5 to 6 cm above the ground plane.
6. Integrally regulated alternator shall be located within a screen room. Alternator location, with regard to screen room, is optional for externally regulated alternators.
7. If not specified otherwise, tests will be performed with and without bonding of the negative bus to the ground plane.
8. Cable shall be routed flat and straight with a length of at least two meters.
9. The EMI measurement instrument chassis power ground shall be broken at the isolation transformer to prevent circulation of rf ground currents in the test equipment.
10. Dashed lines represent an alternator with integral regulator.
11. Provided the other test set up requirements are met, the locations of the batteries and load bank, with respect to the screen room and the extent of their EMI shielding, is optional.

WARNING: Be sure all test instrumentation is properly bonded to the ground plane before applying power to prevent a potential shock hazard to personnel.

Figure 11A. Notes for Figures 11 and 18.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:24Z
Check the source to verify that this is the current version before use.



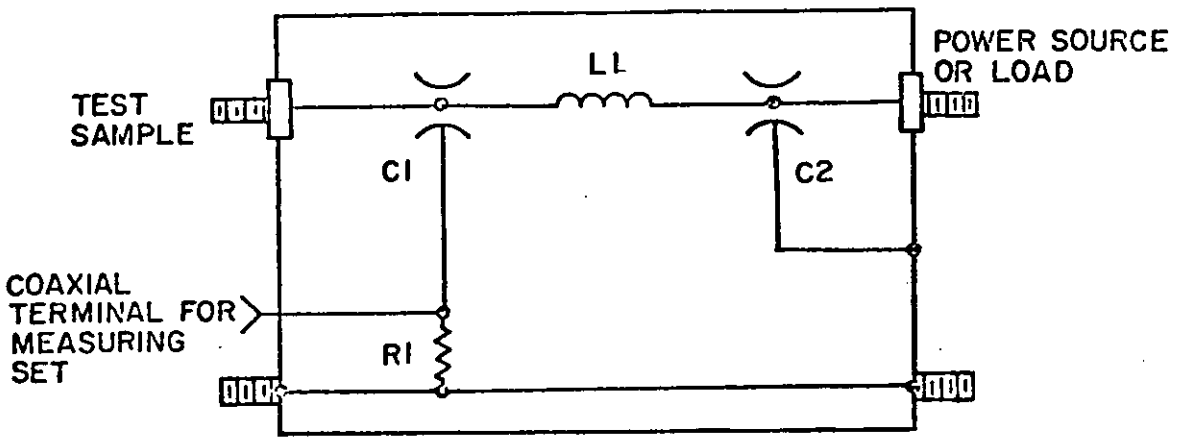
Z_1	Impedance presented to the equipment when connected for measurements.
Z_2	Impedance of the power source used.
BP	Heavy duty binding posts (Mfr. Standard Electric Time Co.).
PR_1	Power receptacle, 115 V, 15 A (3 wire polarized, twist lock, male base).
PR_2	Power receptacle, 115 V, 15 A (3 wire non-polarized, "U" shaped grounding slot).
NC	Type "N" connector (UG-58/U) panel mounting.
L	Coil - 26 turns of No. 6, 600 V insulated wire (stranded), wound on 5-1/2 dia coil form.
CASE	17-1/2 x 17-1/2 x 8-3/4 brass (divided in two sections by a brass plate 17-1/2 x 8-5/8 x 1/16 thick).

NOTES:

1. Dual line stabilization network consists of two networks as shown above.
2. The values given for the component parts of the network are nominal. Regardless of the construction or deviation from nominal values, the network must have an impedance within 20 percent of that given in figure 15.
3. All dimensions are expressed in inches.

FIGURE 12. A network of the dual line impedance stabilization network (low frequency).

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:24Z
Check the source to verify that this is the current version before use.



ENCLOSURE DATA: 0.063 in (14 GA B&S) aluminum, suggested size 9-3/8 in x 4 in x 4 in.
FORM DATA: 5-1/4 inch length, 3 inch dia (OD), 0.125 inch wall, drill 3/8 inch hole 7/16 inch from each end.
WIRE DATA: AWG No. 6, 600 V, 0.310 inch dia (OD).
COIL DATA: L1 - 5 microHenries, 13 turns, single layer, 4 inch winding length.
CAPACITOR: C1 shall be mounted on 1 inch insulating block above ground.
CAPACITOR DATA: C1 - 0.1 μF, 600 V DC, feedthrough.
C2 - 1 μF, 600 V DC, feedthrough.
RESISTOR DATA: R1 - 5000 ohms, 5 W, carbon.

NOTES:

1. The values given for the component parts of the network are nominal. Regardless of the construction or deviation from nominal values, the network must have an impedance within 20 percent of that given in figure 16.
2. Connecting leads to condensers and resistors should be as nearly as possible to zero length.
3. Networks may also be constructed having a one ohm series resistor between the line and capacitor C2. This 1 ohm resistor shall be made up from ten ohm, one watt compostion resistors.
4. The data given in this figure is suitable for the construction of 50 ampere networks. Larger current-carrying networks may be constructed by increasing both the wire size given for the coil and the size of the overall enclosure.
5. The 50 ohm transmission line should be extended within the enclosure right up to the location where it connects with capacitor C1.
6. NOTE: The network shall be prominently and permanently marked "WARNING - Shock Hazard - Connect case to earth ground before connecting AC power line."
7. Networks procured prior to the date of this specification, but meeting the impedance requirements of figure 16, may still be used.
8. Each network shall be permanently labeled with the direct current and voltage ratings at 60, 400, and 800 Hz.

FIGURE 13. Powerline stabilization network schematic diagram.

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:24Z
Check the source to verify that this is the currentversion before use.

1 - 5 cm standoffs.
2 - Low impedance bond to ground plane.
3 - Current probe.
4 - Test sample chassis ground.
5 - High side.
6 - Return (neutral line).
7 - DC bond impedance between the ground plane and enclosure shall not exceed 2.5 milliohms.
8 - Line impedance stabilization networks shall be terminated in 50 ohms resistive at Type "N" connector.

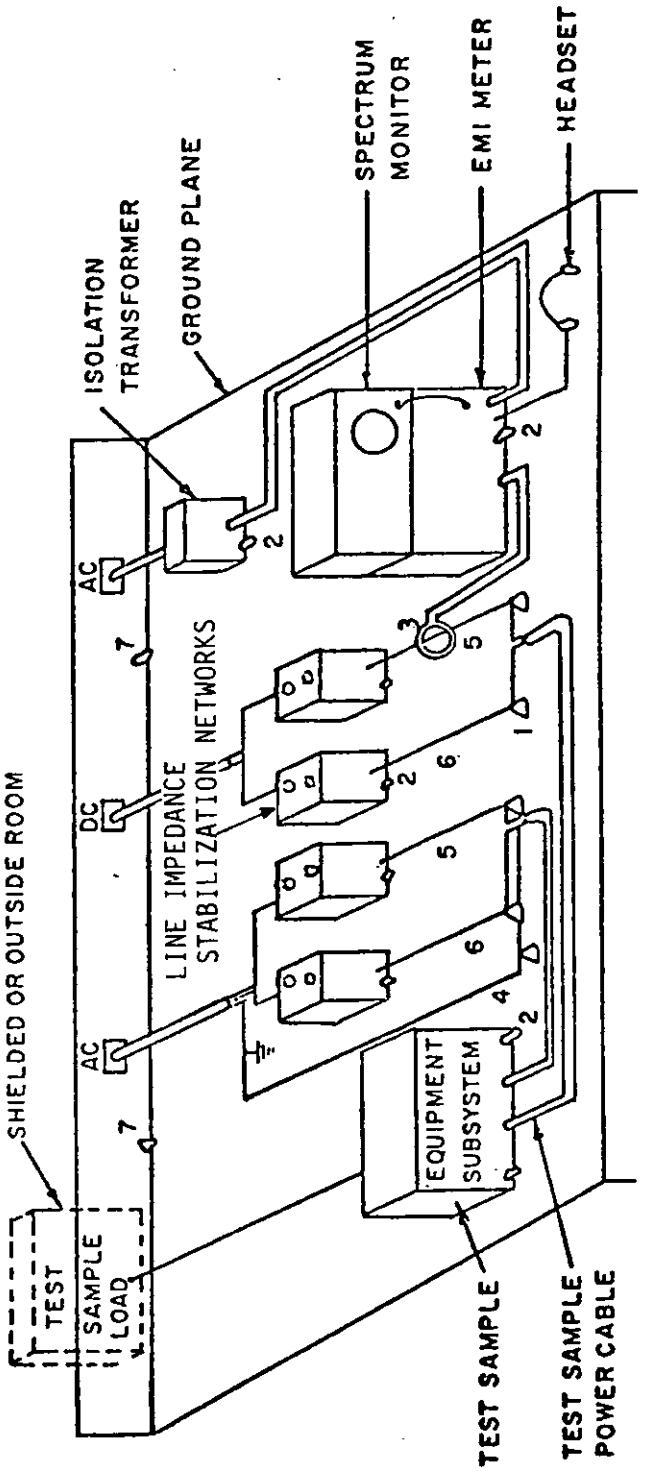


FIGURE 14. Typical test set-up for conducted emission measurements on power leads.

(See figure 14A.)

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:24Z
Check the source to verify that this is the current version before use.

1. The equipment power ground shall be referenced to the ground plane for this test.
2. The separation between cables, leads, and ground plane shall be at least 5 centimeters.
3. The length of power leads from the test sample to the stabilization networks shall be 2 to 3 meters.
4. The length of each power lead between the point of separation and connection to the stabilization network shall be 30 $\pm$ 2 cm.
5. The test sample and EMI instrumentation should derive their power requirements from separate phases of the ac power source, if possible. The purpose of this requirement is to provide isolation between the test sample and measurement instrumentation through the shielded enclosure power line filters.
6. The EMI measurement instrument chassis power ground shall be broken at the isolation transformer to prevent circulation of rf ground currents in the test equipment.
7. Calibration generators and other electronic measurement instrumentation shall not contribute adversely to the power line ambient level. Separate power source phases, filters, reversed line impedance stabilization networks, etc. shall be used as required to provide rf isolation.

WARNING: Be sure all test instrumentation is properly bonded to the ground plane before applying power to prevent a potential shock hazard to personnel.

FIGURE 14A. Notes for Figure 14.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:24Z
Check the source to verify that this is the current version before use.

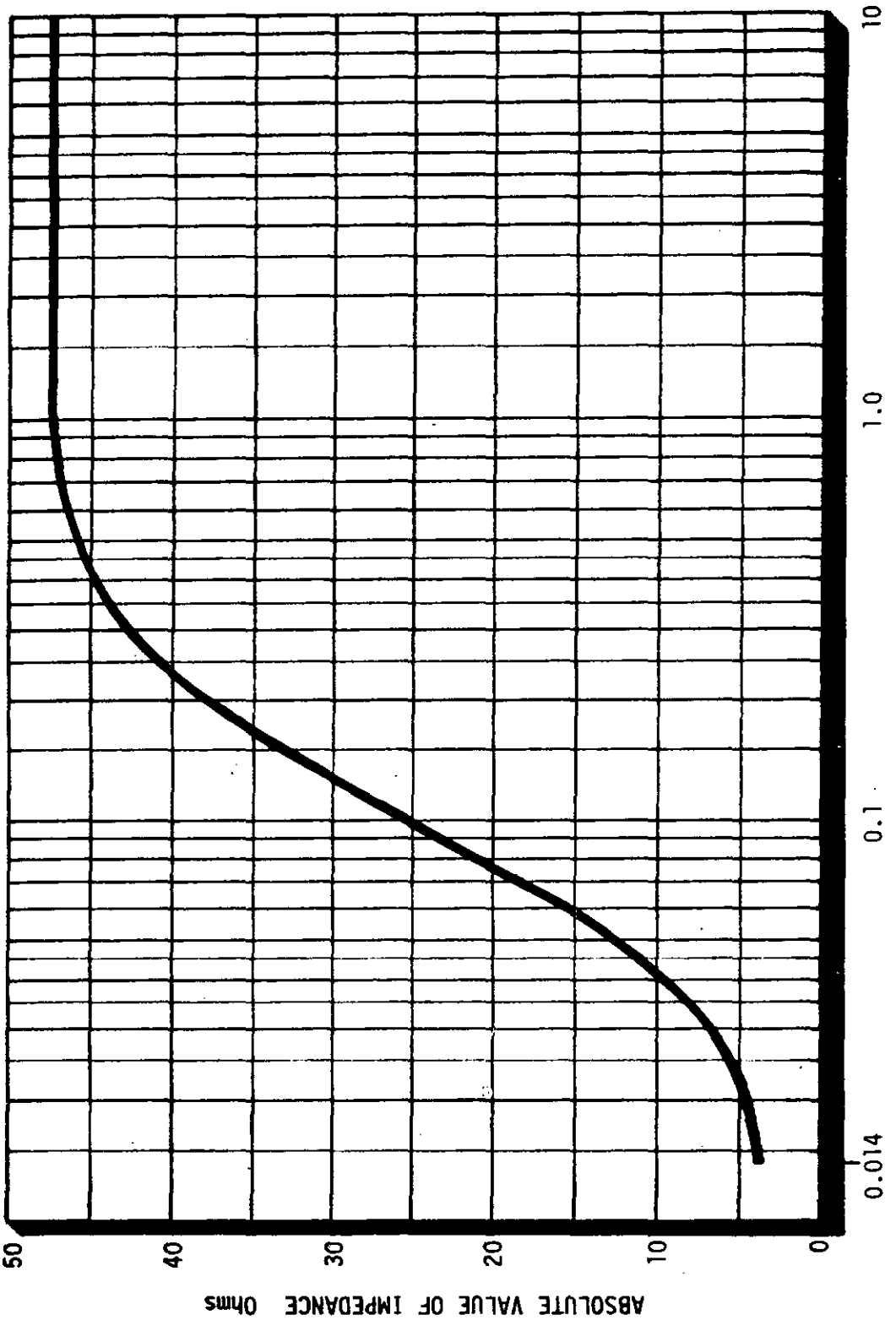


FIGURE 15. Low frequency line impedance stabilization network.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:24Z
Check the source to verify that this is the current version before use.

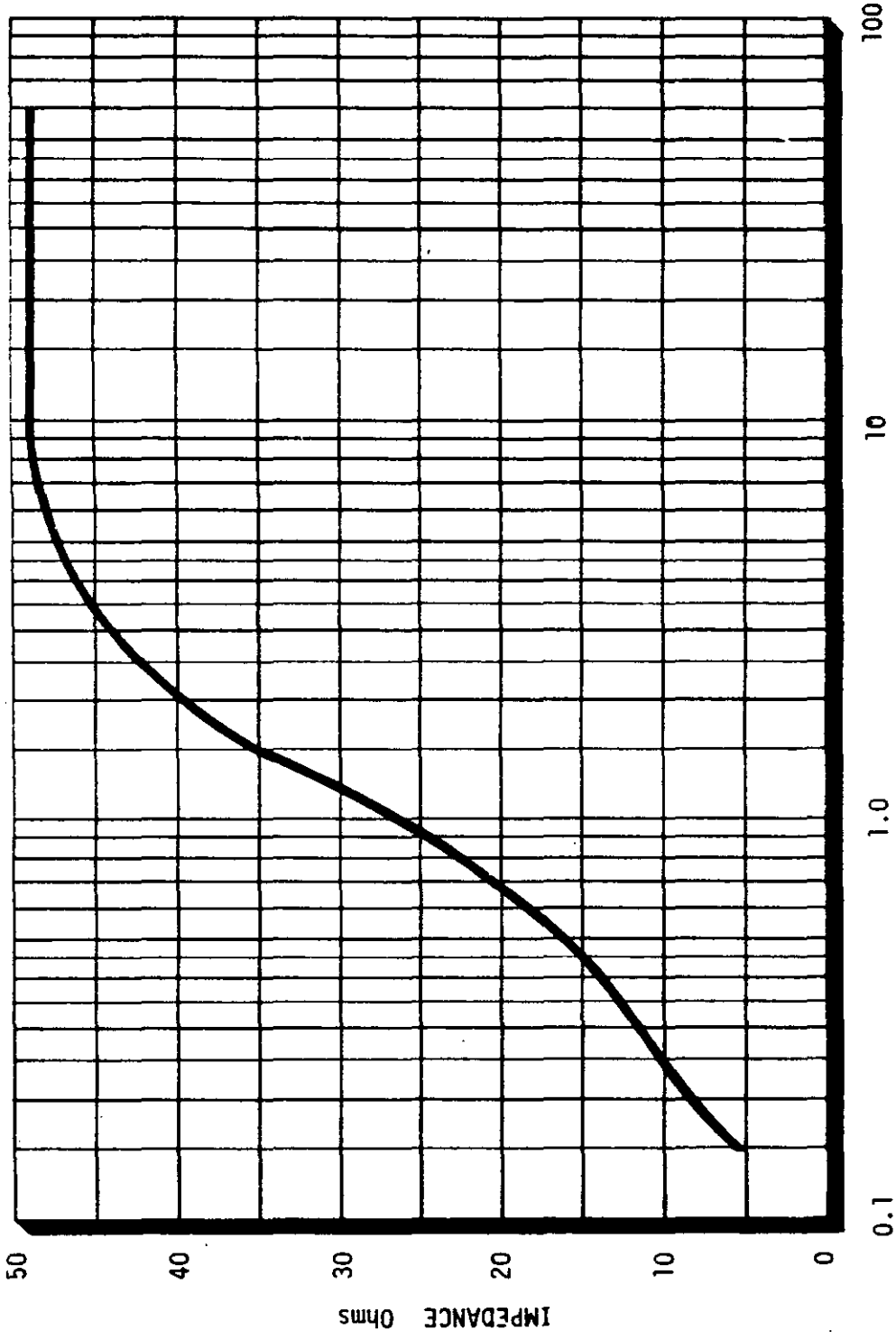


FIGURE 16. Input impedance at test sample terminal of stabilization network with coaxial connector terminated in 50 ohms, power terminal open.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:24Z
Check the source to verify that this is the current version before use.

APPENDIX

80. Method 3 - Emission measurements in high ambient fields (150 kHz - 30 MHz).

80.1 Applicability. The technique specified herein can be used for measuring electromagnetic emissions when high ambient signals prevent the performance of the measurements by the normal methods given in this appendix. This technique shall be used only when authorized by the procuring activity.

80.2 Apparatus. Because of the special techniques specified, the use of the following specific apparatus is required:

a. Raster generator - Honeywell RG-3 or equivalent.
b. Dual beam oscilloscope with P7 phosphor screen - Tektronix type 555 or equivalent.
c. Primary pickup device.
   (1) Antenna unit - Honeywell ARP 3005 or equivalent.
   (2) Current probe - As applicable.
d. EMI meter.
e. Synch pickup probe - LO-205 (part of NF 205) or equivalent.
f. Amplifier (1 kHz - 150 MHz) - Hewlett Packard 461A or equivalent.
g. Signal generators.
   (1) Hewlett Packard Model 606B or equivalent.
   (2) General Radio GR-805D or equivalent.
   (3) Hewlett Packard Model 651B or equivalent.

80.3 Test setup and procedure. The basic test setup is shown on figure 17. The choice of pickup devices shall depend on the ambient noise level and the type and level of the emission being detected and measured. The final test configuration shall conform to the measuring requirements of this appendix.

80.3.1 Setting of operating controls.

80.3.1.1 EMI meter. The input attenuator, rf gain control and audio level control shall be set so that there will not be any overloading or signal clipping.

80.3.1.2 Raster generator. The raster generator accepts detected video from the EMI meter. After amplification in a video amplifier stage, the video is processed by a paraphase amplifier where either normal or inverted signal amplification may be selected. A final video amplifier restores the signal to the proper level for cathode follower output to the Z output jack and the audio stage. Along with the video signal processing, a variable sweep generator produces a sawtooth wave form which can be made to recur at a predetermined rate obtained either externally or internally as chosen by the operator.

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:24Z
Check the source to verify that this is the current version before use.

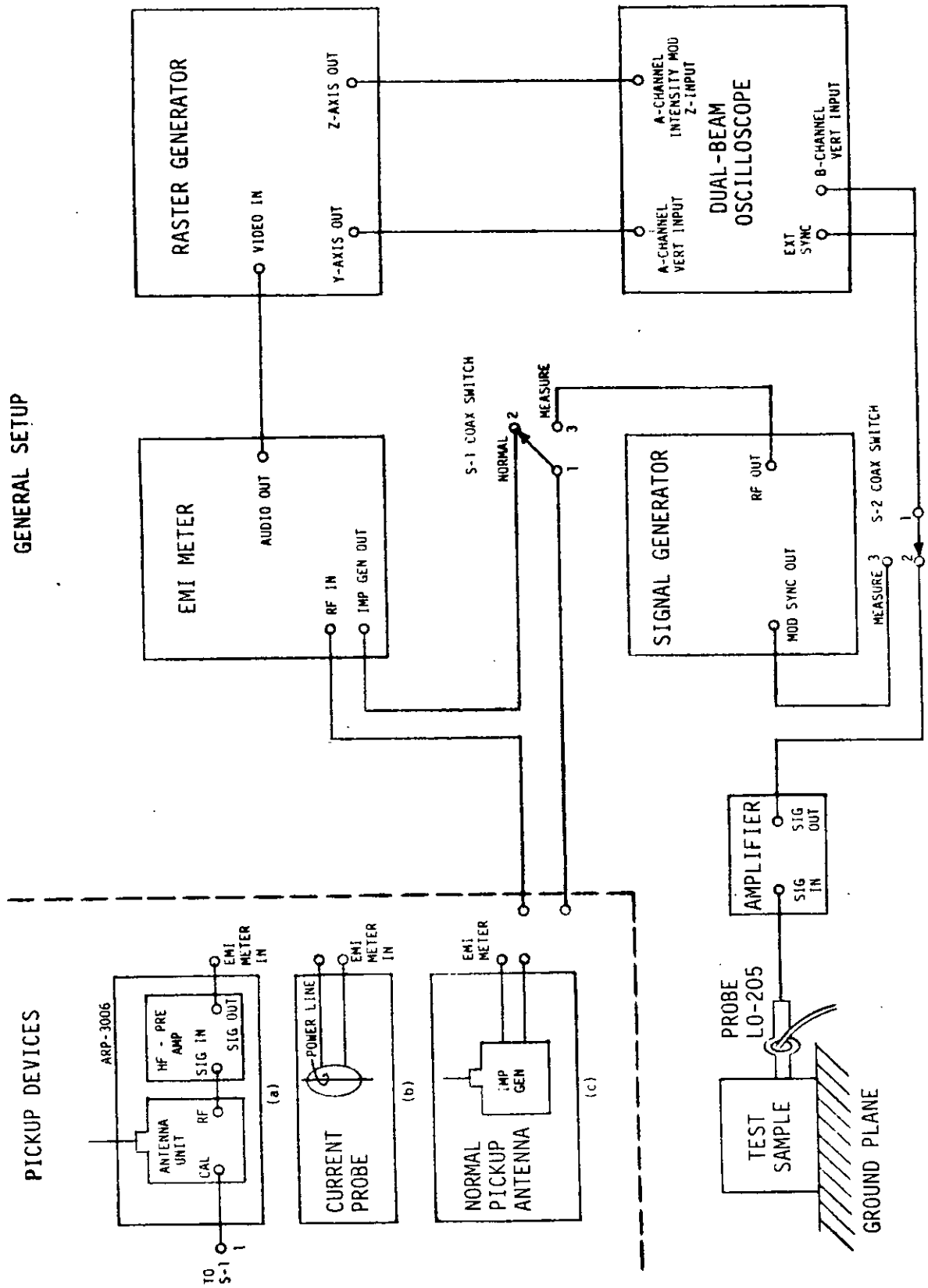


FIGURE 17. Test setup for emission measurements in the presence of high ambient fields.

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:24Z
Check the source to verify that this is the current version before use.

80.3.1.3 Sampling pickup system. The controls shall be set as follows:

a. Probe - Place the probe at the desired point on the test sample to pick up sufficient synch signal.
b. Amplifier H.P. Model 461A - set the gain control for a normal presentation on the oscilloscope. There shall be no clipping or distortion of the synch signal.

80.3.1.4 Dual beam oscilloscope. The controls shall be set as follows:

a. Trigger - Set for external synch.
b. Vertical sensitivity controls - Set for normal presentation on A and B channels. There shall be no clipping or distortion of signals.
c. Other controls - Set to compensate for signal amplitude, shape, and frequency.

80.3.2 Measurement procedures. Measurement procedures shall be as follows:

a. Scan the full frequency range of the lowest tuning band of the EMI meter for the frequency of maximum emission.
b. Adjust the operating controls of the EMI meter, raster generator, dual beam oscilloscope, and signal sampling amplifier to produce the desired display on the dual beam oscilloscope.
c. Adjust the trace on the dual beam oscilloscope so that the vertical sweeps are positioned symmetrically about the zero voltage reference axis.
d. Reduce the channel A intensity on the oscilloscope to the fadeout point.
e. Set the channel B intensity on the oscilloscope to a point where the brightness of the presentation does not obscure the raster presentation on channel A.
f. Readjust the gain and attenuation controls of the EMI meter, raster generator, sampling amplifier, and oscilloscope controls to obtain channel A and B waveforms which do not overload the video amplifiers.
g. Adjust the sweep rate controls on the raster generator and oscilloscope for the desired frame rate.
h. Correlated, repetitive interference pulses should appear as vertical columns of dots or solid vertical lines depending upon the sensitivity setting of the oscilloscope vertical amplifier. This is a means of separating the source interference signal from any high ambient random noise. The interference source signal is thus identified when this correlation is found between the monitor signal and the raster presentation.
i. Adjust the channel A intensity control so that the correlated source interference signal is just visible.

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:24Z
Check the source to verify that this is the current version before use.

j. Turn off the test sample.
k. Place S1 and S2 in the measure position.
l. Tune the signal generator to the same frequency as the EMI meter.
m. Place the signal generator modulation control at 400 Hz, 30 percent modulation.
n. Without resetting operating controls of the EMI meter, raster generator and oscilloscope, set the rf output of the signal generator so that the writing on the oscilloscope equals the intensity as in step d.
o. The induced voltage of the emission at the antenna shall equal the output voltage of the signal generator minus antenna injection network loss.

90. Method 4 - Radiated emissions, electric field.

90.1 Applicability. This method is applicable for the measurement of radiated emissions from all units, cables (including control and power cables), and their interconnecting wiring.

90.2 Frequency range of test. Broadband emissions - 14 kHz to 1000 MHz.

90.3 Apparatus.

a. Test antennas (see 40.1).
b. EMI meter (see table VI).

90.4 Test setup. The basic test setups shall be as shown in figure 18. Test sample antenna terminals, if any, shall be connected to shielded dummy loads.

90.4.1 Procedure.

a. Probe the test sample as indicated in 40.5 and 40.6 to locate the points of maximum radiation from the test sample.
b. Select and position the test antennas as indicated in 40.5 and 40.6 at a test distance of 3 feet (1 meter). In the frequency range of 0.030 to 10.0 GHz, position the linearly polarized antennas so as to make both vertical and horizontal measurements. Bond the counterpoise of the 41 inch rod antenna in accordance with 40.2.
c. For each antenna, scan the applicable frequency range of this test with the EMI meter and take measurements as required.

NOTES:

Errors due to reflections within the shielded enclosure can be reduced by use of absorptive materials lining the walls or by portable panels of such materials placed around the test setup.

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:24Z
Check the source to verify that this is the current version before use.

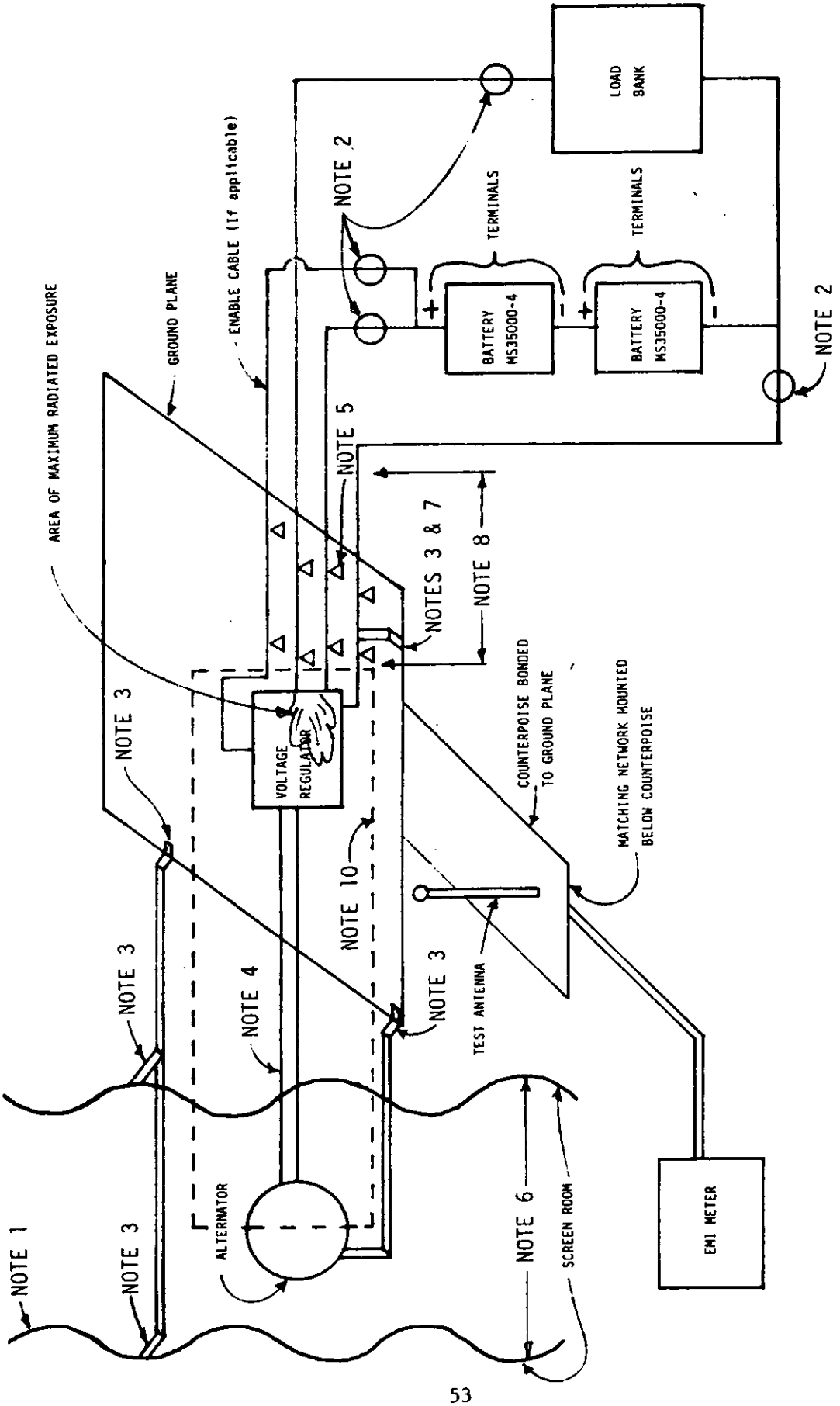


FIGURE 18. Test set-up for Method 4.

(See figure 11A for explanation of notes.)

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:24Z
Check the source to verify that this is the current version before use.

INSTRUCTIONS: In a continuing effort to make our standardization documents better, the DoD provides this form for use in submitting comments and suggestions for improvements. All users of military standardization documents are invited to provide suggestions. This form may be detached, folded along the lines indicated, taped along the loose edge *(DO NOT STAPLE)*, and mailed. In block 5, be as specific as possible about particular problem areas such as wording which required interpretation, was too rigid, restrictive, loose, ambiguous, or was incompatible, and give proposed wording changes which would alleviate the problems. Enter in block 6 any remarks not related to a specific paragraph of the document. If block 7 is filled out, an acknowledgement will be mailed to you within 30 days to let you know that your comments were received and are being considered.

NOTE: This form may not be used to request copies of documents, nor to request waivers, deviations, or clarification of specification requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

(Fold along this line)

(Fold along this line)

DEPARTMENT OF THE ARMY

NO POSTAGE NECESSARY IF MAILED IN THE UNITED STATES

OFFICIAL BUSINESS
PENALTY FOR PRIVATE USE $300

BUSINESS REPLY MAIL
FIRST CLASS PERMIT NO. 12062 WASHINGTON D. C.
POSTAGE WILL BE PAID BY THE DEPARTMENT OF THE ARMY

Commanding Officer
US Army Tank-Automotive
Command
Attn: AMSTA-GSS
Warren, MI 48090

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:24Z
Check the source to verify that this is the current version before use.

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

(TO DETACH THIS FORM, CUT ALONG THIS LINE.)

DD FORM 82 MAR 1426 PREVIOUS EDITION IS OBSOLETE.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:24Z
Check the source to verify that this is the current version before use.